

UC San Diego

Intervening at the Source:

A Data-Centric Approach to Trustworthy AI

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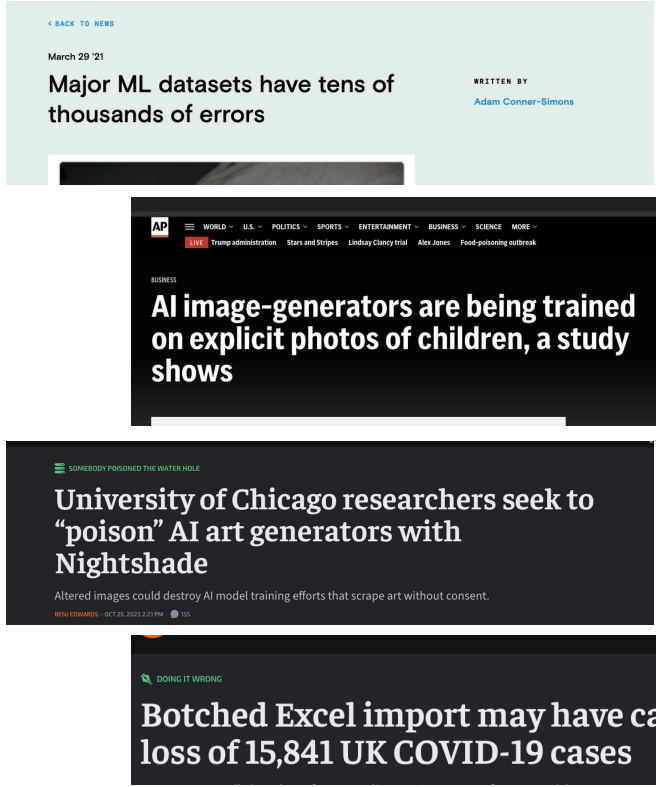
The case for trustworthy AI

- AI increasingly shapes decisions that affect people's lives: who gets considered for a job, who receives credit, and who gets access to health care.
- Greater autonomy raises the stakes. As AI systems act with less human oversight, there are fewer opportunities to catch mistakes before they cause harm.
- We have already seen AI systems fail in consequential ways: they have favored one group over another in hiring, contributed to wrongful arrests through facial-recognition errors, fabricated legal citations, reproduced sensitive training text, and become less safe after fine-tuning.
- This motivates trustworthy AI: systems that are accurate, reliable, and robust in their intended use, while also being fair, privacy-preserving, safe, secure, transparent, and accountable.



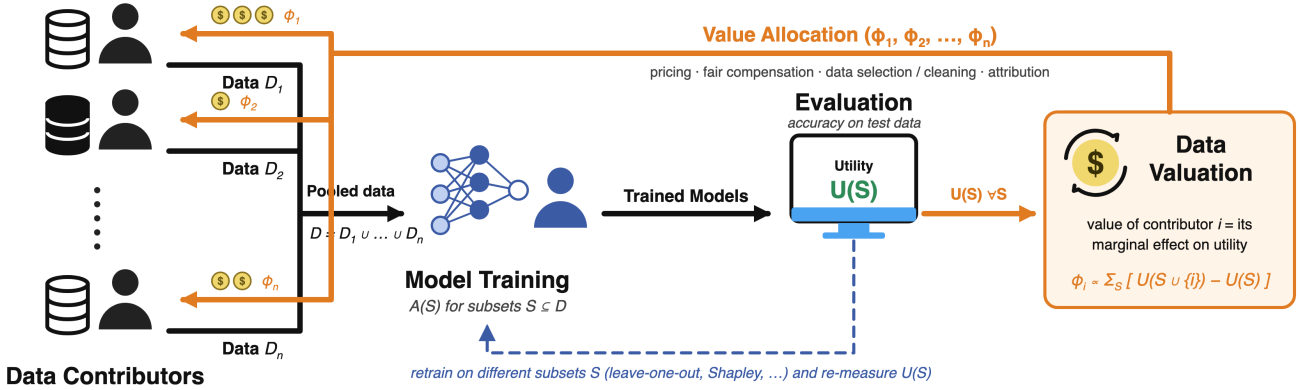
Data is a powerful point of intervention

- Data shapes what models learn: the capabilities they develop, the patterns they rely on, and the mistakes they make.
- Data problems can undermine learning and evaluation: biased samples, class imbalance, missing or corrupted values, incorrect or inconsistent labels, inconsistent formats or units, duplicates, outdated records, train–test leakage, and poisoned or harmful content.
- My research focuses on understanding how data shapes learning and developing principled methods to improve the data that learning systems rely on.



Data valuation

Data valuation assigns a score to a data contribution based on how much it helps or hurts a chosen objective.



- The unit can be one example, a batch, or a whole dataset.
- The objective sets the score: model performance or a data-level measure.
- A contribution can be negative. The same data can have different value for different tasks.

What data valuation enables

Data selection

Choose a compact coreset for efficient training.

Data reweighting

Use data values to adjust how strongly each example contributes to training.

Data minimization

Retain less data while limiting loss in task performance.

Data acquisition

Choose which data to collect or buy next.

Active learning

Spend the labeling budget on examples expected to help most.

Domain adaptation

Prioritize source examples that help on the target domain.

Data pricing

Use a dataset's contribution to help determine its price.

Explanation and debugging

Trace predictions and failures back to influential training examples.

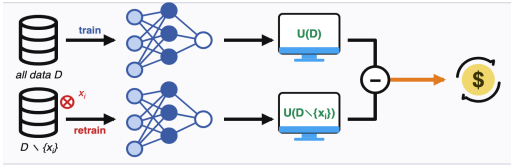
Data cleaning

Find errors and possible poisoning for inspection and repair.

Existing model-dependent valuation methods

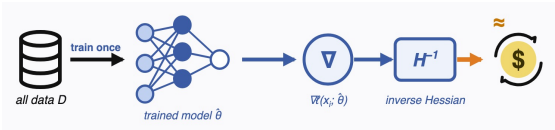
Both approaches tie data values to the chosen model.

Retraining-based valuation



- Compare utility after training with and without a data contribution.
- Repeated training can be costly across examples or data subsets.

Gradient-based methods



- Use model derivatives to estimate how a data contribution affects utility.
- Less retraining, but accuracy depends on the approximation assumptions.

Can we value data efficiently without choosing a downstream model, while still capturing its usefulness for different learning tasks?

What you will be able to explain

- Specify a valuation objective and its evaluation population.
- Calculate removal value, local influence and coalition credit—and explain their differences.
- Choose and test a responsible data intervention.

Begin with the valuation contract

Contract	What must be specified
Player	One record, document, person or source
Learner A	Model, optimizer, initialization and seeds
Utility U	Metric and higher-is-better convention
Population E	Whose outcomes count, including groups
Baseline U(∅)	No data, or a pretrained base model

- Compare scores only when the valuation contracts match.
- A rare record can look low-value when its group is absent from the evaluation population.

Utility connects a subset to an outcome

Stage	Data used	Purpose
Training	Candidate subset S	Fit model weights
Selection	Development examples	Choose data or settings
Final testing	Untouched evaluation examples	Measure generalization

$$U(S) = \mathrm{Eval}_E(A(S))$$

S: training subset · A: fixed training procedure · E: evaluation population · U: higher is better

- Train on subset S, then evaluate the resulting model.
- Keep the evaluation rule fixed when comparing subsets.
- Reserve final testing from adaptive selection.

Value is one question among several

Question	Example of a distinction
Contribution	Correct duplicate: little extra utility
Quality	A low score does not prove a label error
Representation	Rare group can be missing from evaluation
Rights	Consent and compensation need their own rules

- Value asks about contribution to a specified objective.
- Quality asks whether data are accurate, suitable and well documented.
- Consent, privacy, fairness and compensation require additional judgments.

Three different contribution questions

Score	What changes?	Background context
Leave-one-out	Remove a whole player	All other players remain
Classical influence	Infinitesimal loss weight	Near one fitted solution
Shapley	Add player to a coalition	Average across arrival orders

- Removal: what changes if this player leaves the full dataset?
- Influence: what changes under a tiny reweighting near a fitted solution?
- Shapley: what does this player add to different subsets of players (coalitions)?

Leave-one-out holds the other players fixed

Run	Training input	Measured utility
Full fit	D	U(D)
Refit	D without player i	U(D without i)

$$v_i^{\mathrm{LOO}} = U(D) - U(D \setminus \{i\})$$

D: full dataset · i: one player · U: fixed higher-is-better utility

- Compare full-data utility with utility after removing player i.
- Positive value means removal hurts the chosen utility.
- One background coalition can hide redundancy.

Illustration: three neighbors, one query

Player	Distance to query	Label
A	1	1
B	2	0
C	3	1

- Predict the label of one query whose true label is 1.
- Use the closest available neighbor; the empty set has utility 0.
- Utility is 1 for a correct prediction and 0 otherwise.

Compute utility for each coalition

S	∅	A	B	C	AB	AC	BC	ABC
U(S)	0	1	0	1	1	1	0	1

- A wins whenever it is present; otherwise B beats C.
- AB and AC predict 1; BC predicts 0.
- These eight utilities define the entire game.

Try it: who matters to the full dataset?

S	∅	A	B	C	AB	AC	BC	ABC
U(S)	0	1	0	1	1	1	0	1

- Full-data utility is U(ABC) = 1.
- What are the removal values of A, B and C?
- Does a zero removal value mean an example is useless?

Worked feedback: removal sees one context

Removed player	Remaining coalition	Utility after removal	Value
A	BC	0	1 – 0 = 1
B	AC	1	1 – 1 = 0
C	AB	1	1 – 1 = 0

- A: 1 – U(BC) = 1. B: 1 – U(AC) = 0. C: 1 – U(AB) = 0.
- C is redundant beside A, yet useful when A is absent.
- Leave-one-out is an intervention at the full dataset.

Replace repeated refitting with a local question

$$R(\theta) = \frac{1}{n}\sum_{j=1}^{n} \ell_j(\theta), \qquad \hat{\theta}_\epsilon = \arg\min_\theta \{R(\theta) + \epsilon \ell_i(\theta)\}$$

n: training examples · R: average training loss · ℓ_i: chosen example loss · ε: additive weight · $\hat{\theta}_\epsilon$: perturbed fit

- Fit parameters θ by minimizing average training loss.
- Add a small weight ε to one example's loss.
- Estimate how a query loss changes near the fitted solution.

Influence combines gradients and curvature

$$H\frac{d\hat{\theta}}{d\epsilon} + g_i = 0$$

$$\frac{d\hat{\theta}}{d\epsilon} = -H^{-1}g_i$$

$$\frac{d\ell_q}{d\epsilon} = g_q^\top \frac{d\hat{\theta}}{d\epsilon} = -g_q^\top H^{-1} g_i$$

g_i/gq: training/query loss gradients · H: risk Hessian · ε: additive weight · T: transpose · differentiable stable fit, invertible H · All derivatives at ε = 0 (the original fit)

- The training gradient tells us how this example pushes the fitted weights.
- The inverse Hessian accounts for local curvature.
- Positive query-loss influence means harmful upweighting.

Illustration: a model that predicts one constant

$$R(\theta) = \frac{1}{3} \sum_{y \in \{0,1,2\}} \frac{1}{2}(\theta - y)^2$$

$$R\prime(\theta) = \theta - 1 = 0 \;\Rightarrow\; \hat{\theta} = 1, \quad H = R\prime\prime(\theta) = 1$$

θ: constant prediction · R: average training loss · query target yq = 2

- Training targets are 0, 1 and 2; predict the same number θ for every input.
- Squared loss is one half of the prediction error squared.
- The fitted mean is θ = 1; the query target is 2.

Work through the local derivative

Ingredient at θ = 1	Calculation	Value
Training gradient	θ – 0	+1
Query gradient	θ – 2	–1
Curvature	R″(θ)	1

$$\frac{d\theta}{d\epsilon} = -1, \qquad \frac{d\ell_q}{d\epsilon} = (-1)(-1) = +1$$

Perturbed objective: R(θ) + ε · ½(θ – 0)²

- At θ = 1, the target-0 training gradient is +1 and the query gradient is –1.
- The parameter derivative is –1; the query-loss derivative is +1.
- Upweighting target 0 hurts this target-2 query locally.

A derivative is not exact retraining

$$\hat{\theta}_{-0} = \frac{1+2}{2} = \frac{3}{2}, \qquad \ell_q(\hat{\theta}) = \frac{1}{2} \longrightarrow \ell_q(\hat{\theta}_{-0}) = \frac{1}{8}$$

Full fit: targets {0,1,2}, $\hat{\theta} = 1$ · query target 2 · loss $\frac{1}{2}(\theta - 2)^2$ · additive deletion $\varepsilon = -1/3$

- Deleting target 0 corresponds to $\varepsilon = -1/3$ in this objective.
- First-order θ change: +1/3. Exact refitting change: +1/2.
- First-order query-loss change: −1/3. Exact change: −3/8.

Try changing whose outcome matters

$$\frac{d\ell_q}{d\epsilon} = -(\theta - y_q)H^{-1}g_i$$

Given θ = 1, H = 1, g_i = +1; new query target yq = 0

- Return to the full-data fit θ = 1; keep H = 1 and the target-0 gradient +1.
- Change the query target from 2 to 0.
- What happens to the sign of target-0 influence?

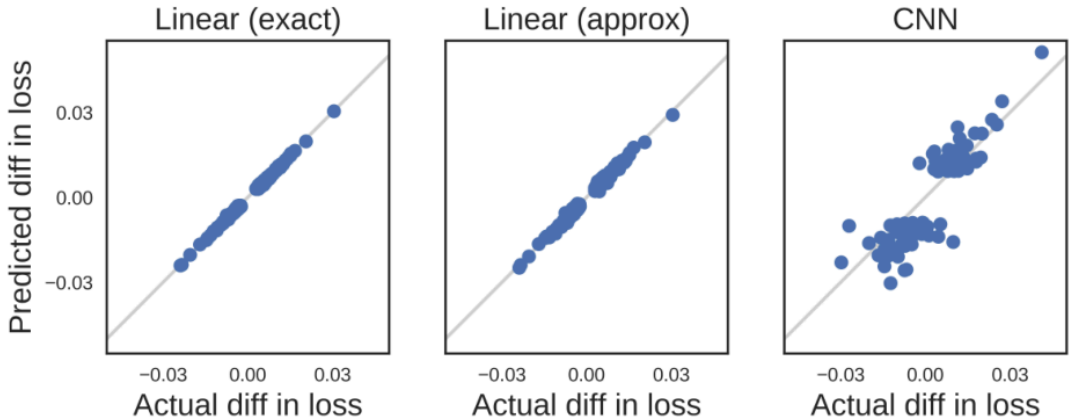
Feedback: value changes with the query

Query target	Query gradient at θ = 1	Upweight-loss derivative
2	−1	+1: locally harmful
0	+1	−1: locally helpful

- The query gradient is now +1, so the loss derivative is −1.
- Upweighting target 0 helps the target-0 query locally.
- The record did not change; the evaluation target did.

Does predicted influence track actual change?

Koh & Liang · ICML 2017



- MNIST: compare predicted and actual removal effects for one wrongly classified query.
- Agreement in these models supports a local approximation—not universal fidelity.

A different question: credit across contexts

Context before C arrives	Utility before	Utility after adding C
No player	0	1
A already present	1	1
B already present	0	0

- A player can help before a redundant substitute arrives.
- Shapley averages that player's marginal utility across coalitions.
- The training procedure and utility still define the game.

The marginal contribution depends on its predecessors

$$\phi_i = \sum_{S \subseteq D \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \left[U(S \cup \{i\}) - U(S) \right]$$

D: n players · S: predecessors of i · |S|: coalition size · factorial weight counts arrival orders · ϕ_i: Shapley credit

- In one random order, let P_i be the players before i.
- Credit is $U(P_i \cup \{i\}) - U(P_i)$.
- Average this quantity over all arrival orders.

Trace the order B → C → A

Query label 1 · A(distance 1,label 1), B(2,0), C(3,1) · nearest neighbor predicts · $U(\varnothing) = 0$

- B arrives: $U(B) - U(\varnothing) = 0$.
- C arrives: $U(BC) - U(B) = 0$.
- A arrives: $U(ABC) - U(BC) = 1$.

Try it: average all six arrival orders

Order	ΔA	ΔB	ΔC
ABC	1	0	0
ACB	1	0	0
BAC	1	0	0
BCA	1	0	0
CAB	0	0	1
CBA	1	−1	1

- Each row gives marginal contributions in A, B, C order.
- Average each player's column to obtain its Shapley credit.
- Compare the answer with removal values (1, 0, 0).

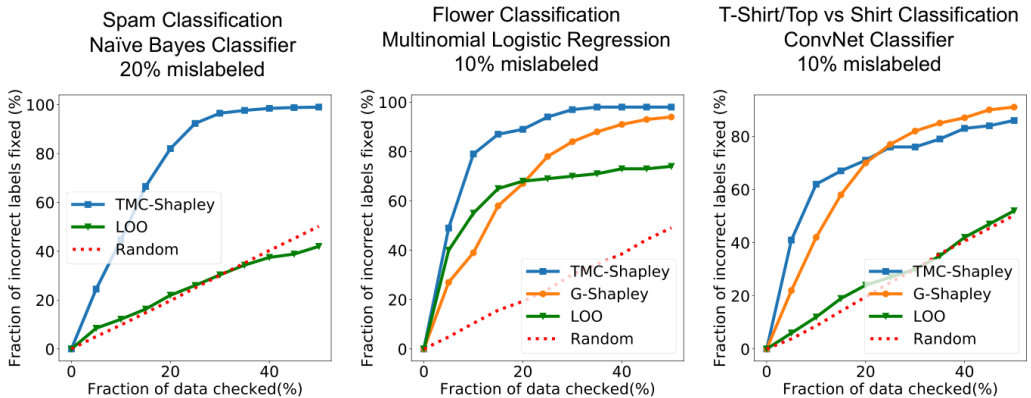
Feedback: coalition credit reveals the redundancy

Order	ΔA	ΔB	ΔC
ABC	1	0	0
ACB	1	0	0
BAC	1	0	0
BCA	1	0	0
CAB	0	0	1
CBA	1	−1	1

- Shapley credit is A = 5/6, B = −1/6, C = 1/3.
- C gets credit for useful early-arrival contexts.
- B hurts this query in one order without being mislabeled.

Can low-value rankings help a label audit?

Ghorbani & Zou · ICML 2019



- Inspect training labels in the order suggested by each method.
- The curve measures recovered corruption versus the fraction checked.
- This supports prioritizing an audit, followed by verification.

Coalition averaging creates a computational problem

$$\widehat{\phi}_i = \frac{1}{M} \sum_{m=1}^{M} \left[U\left(P_i^{\pi_m} \cup \{i\}\right) - U\left(P_i^{\pi_m}\right) \right]$$

M: sampled orders · πm: sampled arrival order · $P_i^{\pi m}$: predecessors of i · each utility may require training

- Exact Shapley can require utilities for exponentially many subsets.
- Sampling arrival orders estimates the average marginal contribution.
- Truncation and approximations trade computation for error.

A special learner can make exact credit easier

Jia et al. · PVLDB 2019

Step	Why work can be shared
Sort neighbors for one query	Every coalition draws from this ordering
Apply the KNN-specific recurrence	Adjacent ranks share marginal calculations
Average over evaluation queries	The evaluation set defines the final value

$$U_q(S) = \frac{1}{K} \sum_{j=1}^{\min(K,|S|)} \mathbf{1}\{y_{\alpha_j(S)} = y_q\}$$

K: fixed neighbor count · $\alpha_j(S)$: j-th nearest member of S · y: labels · missing neighbors contribute 0

- Exact O(n log n) computation is available for its specified K-nearest-neighbor utility.
- This is not an exact algorithm for arbitrary neural-model Shapley.

Try it: whose evaluation population counts?

$$U(S) = 0.8\,\mathbf{1}\{A \in S \text{ or } B \in S\} + 0.2\,\mathbf{1}\{C \in S\}$$

M, R: two domains · A/B: duplicate M coverage · C: R coverage · illustrative utility · 1{condition} = 1 if true, 0 otherwise

- A and B each cover domain M; C covers domain R.
- Utility gives 80% weight to M coverage and 20% to R coverage.
- How should the two duplicate M contributors share credit?

Feedback: change the objective, change the values

Domain weights (M, R)	A	B	C
(0.8, 0.2)	0.4	0.4	0.2
(0.5, 0.5)	0.25	0.25	0.5

- With 80/20 domain weights, Shapley credit is (0.4, 0.4, 0.2).
- With 50/50 domain weights, it becomes (0.25, 0.25, 0.5).
- The allocation inherits the evaluation population.

Player identity also changes the allocation

$$\phi_A + \phi_{A\prime} = 2\left(\frac{0.8}{3}\right) = \frac{8}{15} > 0.4$$

After duplication: A, A′ and B are symmetric M players; C still supplies R

- Suppose A's owner submits an identical extra player A′.
- Three M players now share 0.8; A's owner receives 8/15 in total.
- Record-level symmetry does not guarantee owner-level fairness.

Model-agnostic data valuation

KAIROS: Scalable Model-Agnostic Data Valuation

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NeurIPS 2025



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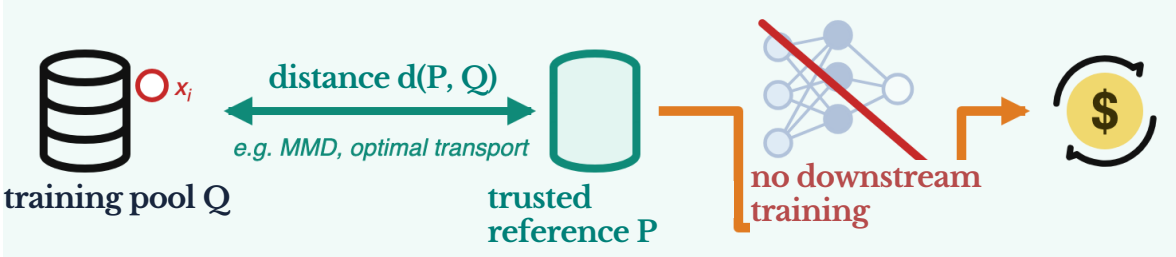
* Equal contribution

THE QUESTION

How can we value data without committing to a particular learning algorithm?

An emerging research direction: data-centric, model-agnostic valuation.

A data-level objective for valuation



Instead of training a model on Q and evaluating it on a clean, representative validation set P, we directly measure the distributional distance d(P, Q).

We value each training point by how it changes the distance between Q and P.

For the same model trained on Q, prior work establishes bounds for many distances, under suitable assumptions:

Target loss ≤ source loss + $C \cdot d(P, Q)$ + additional terms.

From a dataset discrepancy to a point's value



Before removal: Q and fixed P

remove z_i



After removal: Q without the point, same P

$$v_{LOO}(z_i) = d(P, Q_{-i}) - d(P, Q)$$

- Keep the reference P fixed and **remove one point** from the pool Q.
- If discrepancy increases, the point was **helping**. Its value is positive.
- If discrepancy decreases, the point was **hurting**. Its value is negative.
- Exact leave-one-out can require a new discrepancy computation **for every point**.

Which distance?

A value is a change in d(P, Q). Which distance?

- For **disjoint empirical supports**, KL can be infinite and total variation maximal, even when the samples are nearby.
- **Wasserstein and kernel MMD** use relationships between sample locations to capture more informative differences.

Wasserstein-1 and MMD share a test-function view: integral probability metrics.

Both are integral probability metrics

An IPM chooses a **critic** (or test function) whose average score differs most between P and Q.

$$IPM_F(P,Q) = \sup_{f \in F} \left| E_P[f] - E_Q[f] \right|$$

F: allowed critics, $f: X \to \mathbb{R}$

Wasserstein-1 the transport distance with an IPM form

$$W_1(P,Q) = \sup_{Lip(f) \le 1} \left| E_P[f] - E_Q[f] \right|$$

1-Lipschitz critics: scores change no faster than distances between inputs.

Classical exact OT: $O(n^3 \log n)$
+ $O(dn^2)$ to compute pairwise distances.

MMD

$$MMD(P,Q) = \sup_{\|f\|_H \le 1} \left| E_P[f] - E_Q[f] \right|$$

RKHS unit ball: kernel-defined critics with norm at most one.

Gaussian MMD: $O(dn^2)$
Closed-form pairwise kernel sums.

n = number of data points in each dataset; d = data dimension (number of features).

RKHS and MMD: three simple ideas

1. **A kernel** k(x,y) measures similarity.

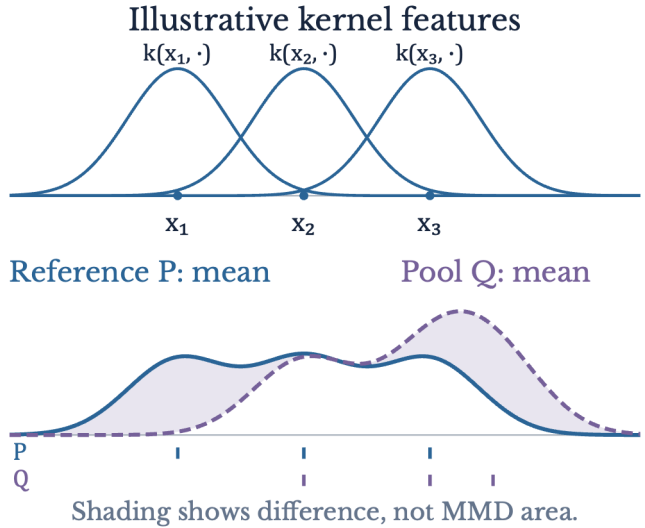
Point feature: $\varphi(x) = k(x,\cdot)$

The RKHS contains weighted sums of these feature functions.

$$\langle k(x,\cdot), k(y,\cdot)\rangle_H = k(x,y)$$

2. **Dataset becomes an average.** Average its feature functions to get its mean embedding.

3. **MMD compares the averages.** Small MMD means the pool looks like the reference.



$$\mathrm{MMD}_k(P,Q) = \|\mu_P - \mu_Q\|_H$$

RKHS norm

$$\mathrm{MMD}_k{}^2(P,Q) = \underbrace{E_{X,X'\sim P}\,k(X,X')}_{\text{within P}} + \underbrace{E_{Y,Y'\sim Q}\,k(Y,Y')}_{\text{within Q}} - 2\underbrace{E_{X\sim P,Y\sim Q}\,k(X,Y)}_{\text{across P and Q}}$$

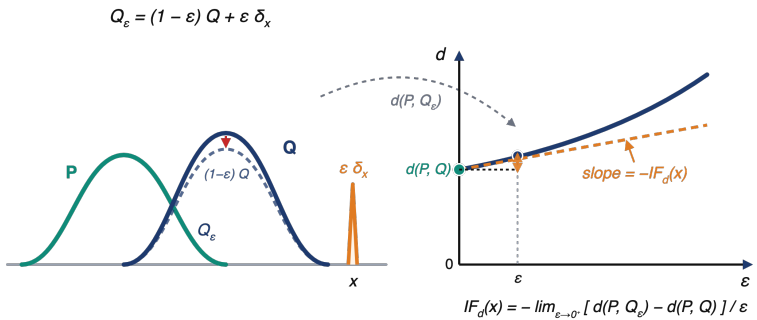
Closed form from **pairwise similarities**, without explicit feature vectors.

First-order influence of a training point

Approximating leave-one-out

Removing one of many points slightly changes the empirical distribution. We estimate the change in distance using its local sensitivity to that point's weight.

$$\mathrm{IF}_d(x; P, Q) = -\frac{d\big(P, (1-\varepsilon)Q + \varepsilon\delta_x\big) - d(P, Q)}{\varepsilon}\Big|_{\varepsilon \to 0^+}, \tag{1}$$



Reweight x and renormalize the other points. Features and labels stay fixed.

What does the sign mean?

Positive influence: more weight on x reduces the discrepancy.

For n > 1 equally weighted points. The removal approximation assumes differentiability across ε = 0.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

IPM sensitivity: two terms

Expanding the influence definition gives the following decomposition.

$$IF_d(x;P,Q) = \underbrace{\left.\frac{\mathbb{E}_Q[f_\varepsilon^\star - f^\star] - \mathbb{E}_P[f_\varepsilon^\star - f^\star]}{\varepsilon}\right|_{\varepsilon\to 0^+}}_{\text{(i) critic-shift term}} + \underbrace{\left(f^\star(x) - \mathbb{E}_Q[f^\star]\right)}_{\text{(ii) point-gap term}}. \tag{3}$$

f*: optimal critic before the nudge. f*ε: optimal critic after the nudge.

Critic-shift term

Loss in the original score gap per unit nudge.

Point-gap term

How much x's score exceeds the pool's average, using the original critic.

Can we compute influence using the original critic?

Reweighting can change which critic best distinguishes P and Q. We must establish when the original critic's point-gap gives the correct influence.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Closed-form MMD influence and its cost

The MMD derivative is just kernel similarity.

Proposition 1. *The influence function for* MMD *as the distance metric is, up to additive and positive multiplicative constants, given by*

$$\mathrm{IF}_{\mathrm{MMD}}(x; P, Q) = \mathbb{E}_{x' \sim P}[k(x', x)] - \mathbb{E}_{x' \sim Q}[k(x', x)]. \tag{6}$$

- Higher similarity to the trusted reference raises the score.
- Higher similarity to the current pool lowers the score.

Closed-form scores: $O(n + m)$ per point; $O(n^2 + nm)$ for all n points.

Recomputing MMD for all n removals: $O(n^3 + n^2m)$ total.

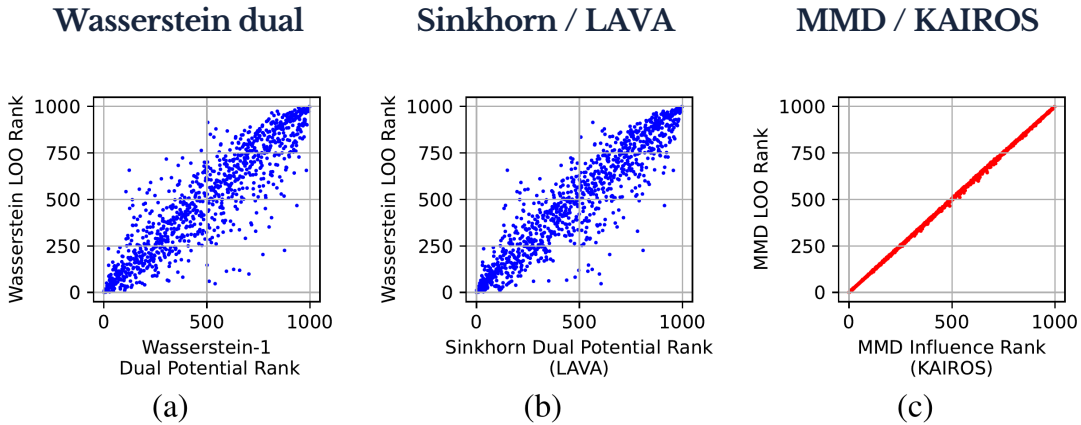
n training points, m reference points. Kernel evaluations; a dense Gaussian kernel adds a factor d.

Zhu, Prashant, Cloninger, Salimi. KAIROS, NeurIPS 2025, Eq. (7) and §3.4. Recompute cost derived from pairwise sums.

Do the scores predict the effect of removing data?

Comparison with prior work: LAVA (Just et al., ICLR 2023).

LAVA values data using sensitivity of a class-wise Wasserstein distance. Here we compare its transport-based scores with KAIROS and actual point removal.



One dot per training point. x: rank by score. y: rank by leave-one-out removal.

In this experiment, top-100 overlap with leave-one-out is **99% for KAIROS** and 60% for LAVA.

Just et al. LAVA: Data Valuation without Pre-Specified Learning Algorithms. ICLR 2023.
Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Properties of KAIROS valuation

- **Symmetry**: equivalent effects on the discrepancy give equal scores.
- **Density separation**: under the proposition's assumptions and a suitable Gaussian bandwidth, scores separate specified density regions.
- **Labels**: a conditional discrepancy compares label behavior between the pool and reference.
- **Streaming**: reuse existing kernel sums and update them for new data.

Proposition 2 (Symmetry). *Let D^{train} and D^{val} be finite samples from distributions Q and P, respectively. If for all subsets $S \subseteq D^{train} \setminus \{x_i, x_j\}$,*

$$\widehat{\mathrm{MMD}}(D^{val}, S \cup \{x_i\}) - \widehat{\mathrm{MMD}}(D^{val}, S) = \widehat{\mathrm{MMD}}(D^{val}, S \cup \{x_j\}) - \widehat{\mathrm{MMD}}(D^{val}, S)$$

then $\widehat{\mathrm{IF}}(x_i) = \widehat{\mathrm{IF}}(x_j)$.

Proposition 3 (Density Separation). *Let P and Q be two probability distributions on $\mathcal{X} \subseteq \mathbb{R}^n$. For any $\epsilon > 0$ and $r > 0$, there exists a Gaussian isotropic kernel k such that for* $\mathrm{IF}(x) = \mathbb{E}_{x' \sim P}[k(x, x')] - \mathbb{E}_{x' \sim Q}[k(x, x')]$, *the following holds*

$$\begin{cases} \mathrm{IF}(x) > 0, & \text{if } P(x') - Q(x') \geq \epsilon \, \forall \, x' \in B(x, r), \\ \mathrm{IF}(x) < 0, & \text{if } P(x') - Q(x') \leq -\epsilon \, \forall \, x' \in B(x, r), \end{cases}$$

where $B(x, r) = \{x' : \|x' - x\| < r\}$.

Proposition 4. *The influence function for* E-MCMD *as the distance metric is, up to additive and positive multiplicative constants, given by*

$$\mathrm{IF}_{\text{E-MCMD}}(x, y; P, Q) = -\|\mu^P_{Y|X}(x) - \phi(y)\|_{\mathcal{H}_\mathcal{Y}} \tag{8}$$

For existing points (x_i, y_i) where $i \leq n_t$, the terms $\mathcal{B}_i$ and $\mathcal{R}_i$ remain unchanged. The training kernel average $\mathcal{A}_i$ can be efficiently updated as:

$$\mathcal{A}_i^{(t+1)} = \frac{1}{n_{t+1} - 1} \left((n_t - 1) \cdot \mathcal{A}_i^{(t)} + \sum_{j=1}^{m} k(x^{\text{train}}_{n_t + j}, x_i) \right)$$

The kernel form supports interpretable scores and efficient updates.

A bound linking discrepancy to transfer loss

Feature and label discrepancies control a loss-gap bound under stated assumptions.

Theorem 1 (Bounding transfer loss (simplified)). *Let $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ be compact metric spaces and $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. Let $L : \mathcal{Z} \to \mathbb{R}$ be a continuous loss function. Then, for some constant c,*

$$\mathbb{E}_{(x,y)\sim Q}[L(x,y)] \leq \mathbb{E}_{(x,y)\sim P}[L(x,y)] + c\Big(\mathrm{MMD}(P_X, Q_X) + \mathbb{E}_{x\sim Q_X}\left[\mathrm{MCMD}_{P,Q}(x)\right]\Big).$$

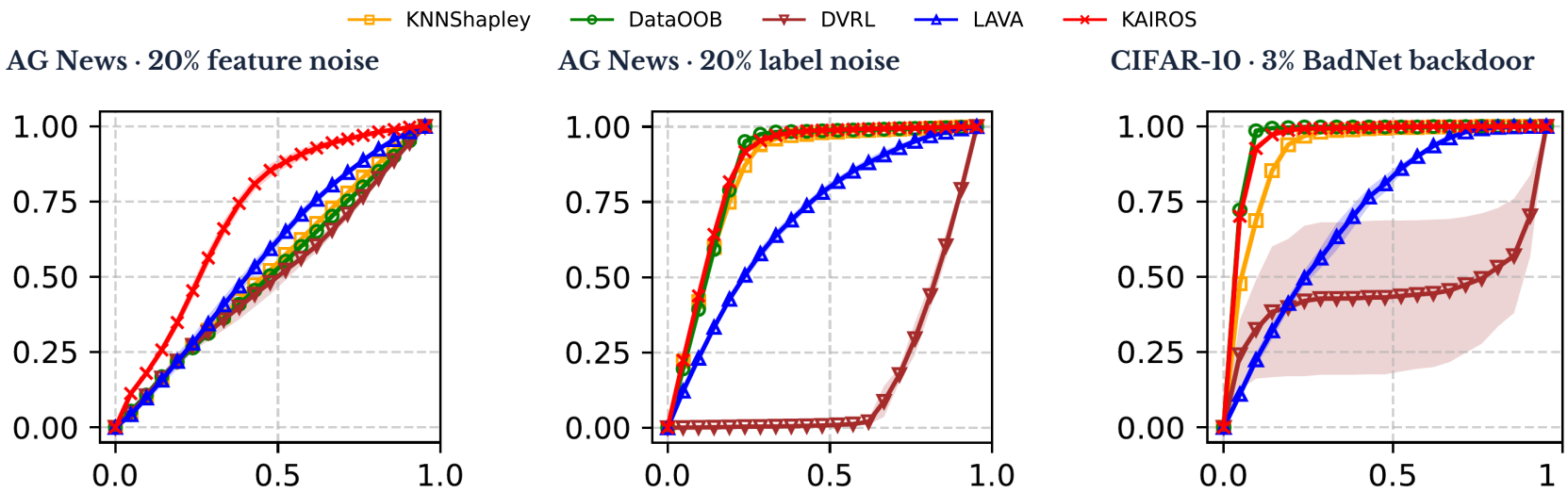
The theorem bounds loss using **feature mismatch and label mismatch**. Reducing those discrepancies tightens the bound's discrepancy term.

Tightens a bound; not a guarantee of better test accuracy.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Finding corrupted examples early

Inspect from the lowest score upward. How much of the corruption is found?



x: fraction inspected. y: fraction of corrupted points found.

BadNets: trigger-marked images with wrong labels (3% poisoned).

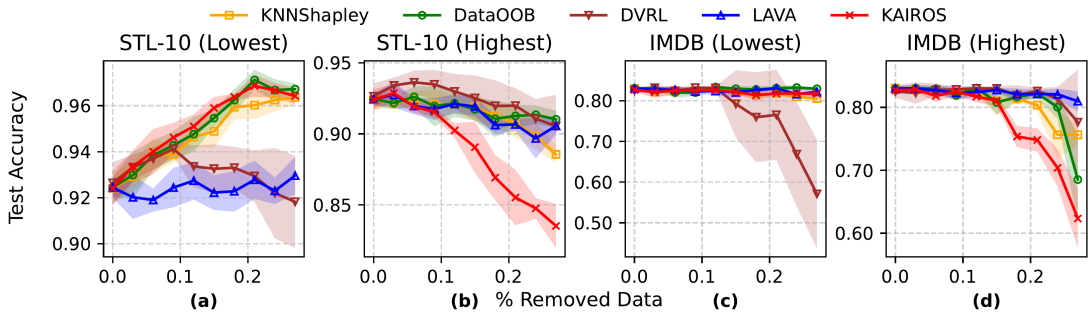
Across eight feature/label-noise settings, KAIROS ranks **first in six and second in two**. The backdoor panel is a separate test.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Testing the ranking and scaling to ImageNet

Does pruning by score behave, and does it scale?

PRUNING BY SCORE · 20% MISLABELED



Test accuracy after removing the lowest-valued (a, c) or highest-valued (b, d) points.

Compare accuracy after removing **low-valued** versus **high-valued** points.

IMAGENET · 1.28M IMAGES

Feature noise	Runtime	AUC
KAIROS	7m56s	0.869
SAVA	1h58m	0.817
Label noise	**Runtime**	**AUC**
KAIROS	7m52s	0.861
SAVA	1h58m	0.484

ResNet-50 features, one A100. AUC of the curve on the previous slide.

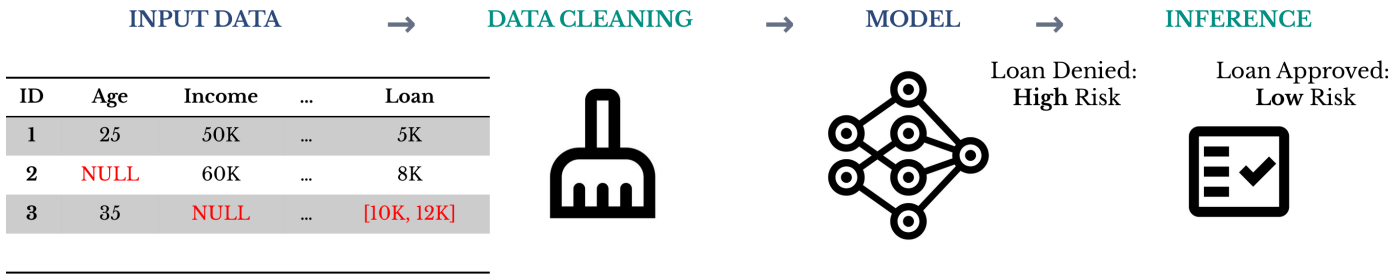
SAVA: a scalable, batched version of LAVA.
Kessler, Le & Nguyen, ICLR 2025.

> On ImageNet features, KAIROS scores 1.28M images in **about 8 minutes**, versus about 2 hours for SAVA, with higher reported detection AUC.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Data selection and prediction uncertainty

- **KAIROS** asks which examples to keep, using a trusted reference to define their value.
- Missing or ambiguous values admit **several plausible repairs**. Different repairs can produce different models.
- **ZORRO** asks how much a prediction can vary across those repairs.



How do **data-cleaning choices** change predictions?

Learning from uncertain training data

Learning from Uncertain Data: From Possible Worlds to Possible Models

Jiongli Zhu[1] Su Feng[2] Boris Glavic[3] Babak Salimi[1]

[1]University of California, San Diego [2]Nanjing Tech University [3]University of Illinois, Chicago

NeurIPS 2024



Jiongli Zhu

UC San Diego

THE QUESTION

How much can a prediction change across plausible repairs of the training data?

UNCERTAINTY-AWARE · SYMBOLIC · CERTIFIED

Possible Worlds Semantics

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Possible Worlds Semantics

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Q: Total income of the three shown rows

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	30	60K	...	8K
3	35	55K	...	11K
...	...	...	...	...

→ Q(D1) = 165K

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

→ Q(D2) = 170K

...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

→ Q(D3) = 170K

...

Answer variability

- Agreement across all represented worlds gives a certain answer. Otherwise, report the range and assess whether it matters for the decision.

Shown range [165K, 170K]

Range of displayed completions

ML from Possible Repairs

Dataset with Quality Issues

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Each plausible repair produces a model. The same input can receive different predictions. Training on every repair separately is infeasible.

Possible Worlds

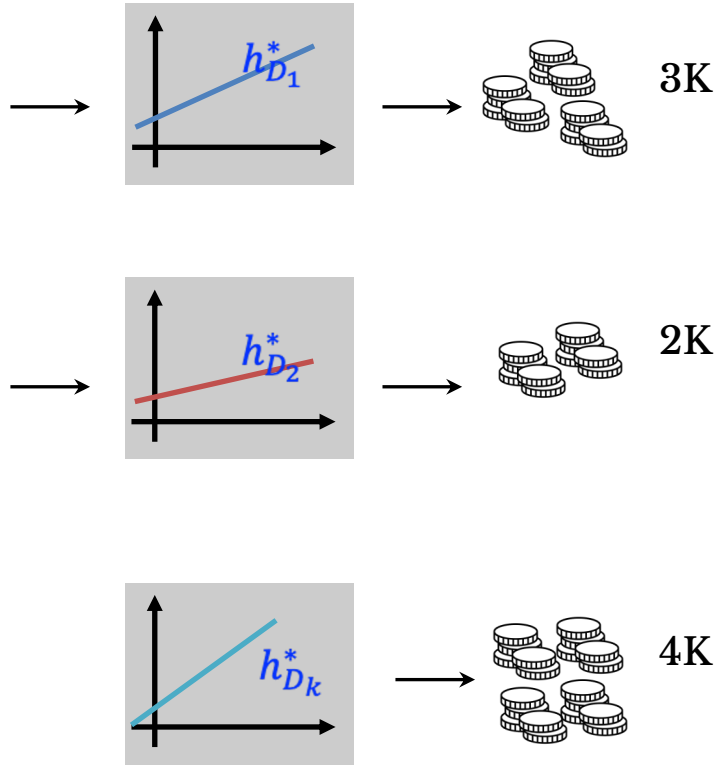
ID	Age	Income	...	Loan
1	25	50K	...	5K
2	30	60K	...	8K
3	35	55K	...	11K
...	...	...	...	...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

Inference

- All represented models agree: certain prediction.
- Otherwise: a range, here 2K–4K.



Compact reasoning over possible models

- Databases represent every possible world compactly: c-tables, probabilistic databases — and relational queries run on the representation.

  [Imieliński & Lipski, JACM 1984] [Sundarmurthy et al., ICDT 2017] [Suciu, Olteanu, Ré & Koch, 2011]
- Reasoning over them is still NP-hard: certain answers are coNP-complete, prediction ranges NP-hard.

  [Abiteboul, Kanellakis & Grahe, TCS 1991] [Zhu, Feng, Glavic & Salimi, ZORRO, NeurIPS 2024]
- No support for numerical optimization: closed under relational algebra, not iterated arithmetic; symbolic gradient descent grows exponentially.
- Abstract interpretation: a representation closed under arithmetic — every training step maps the shape to a shape of fixed size, in polynomial time; sound, precise up to a little slack.

  [Cousot & Cousot, POPL 1977]

ZORRO uses zonotopes to represent uncertainty and its dependencies.

Zonotopes encode possible worlds

Dataset with Quality Issues

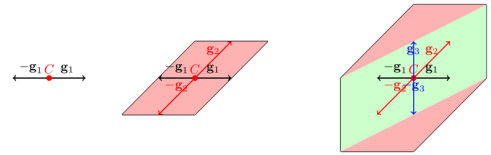
ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Possible Worlds

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	30	60K	...	8K
3	35	55K	...	11K
...	...	...	...	...

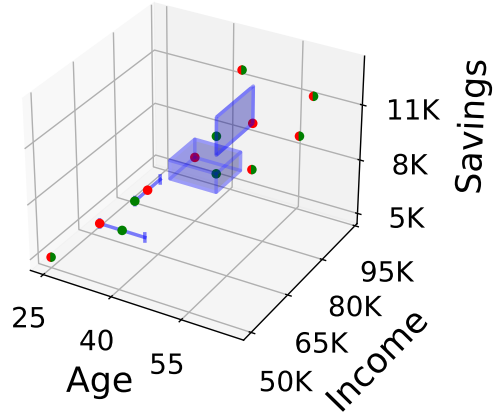
ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...



$c + \varepsilon_1 g_1$ $\quad c + \varepsilon_1 g_1 + \varepsilon_2 g_2$ $\quad c + \varepsilon_1 g_1 + \varepsilon_2 g_2 + \varepsilon_3 g_3$

Each ε_i varies freely in $[-1, 1]$.



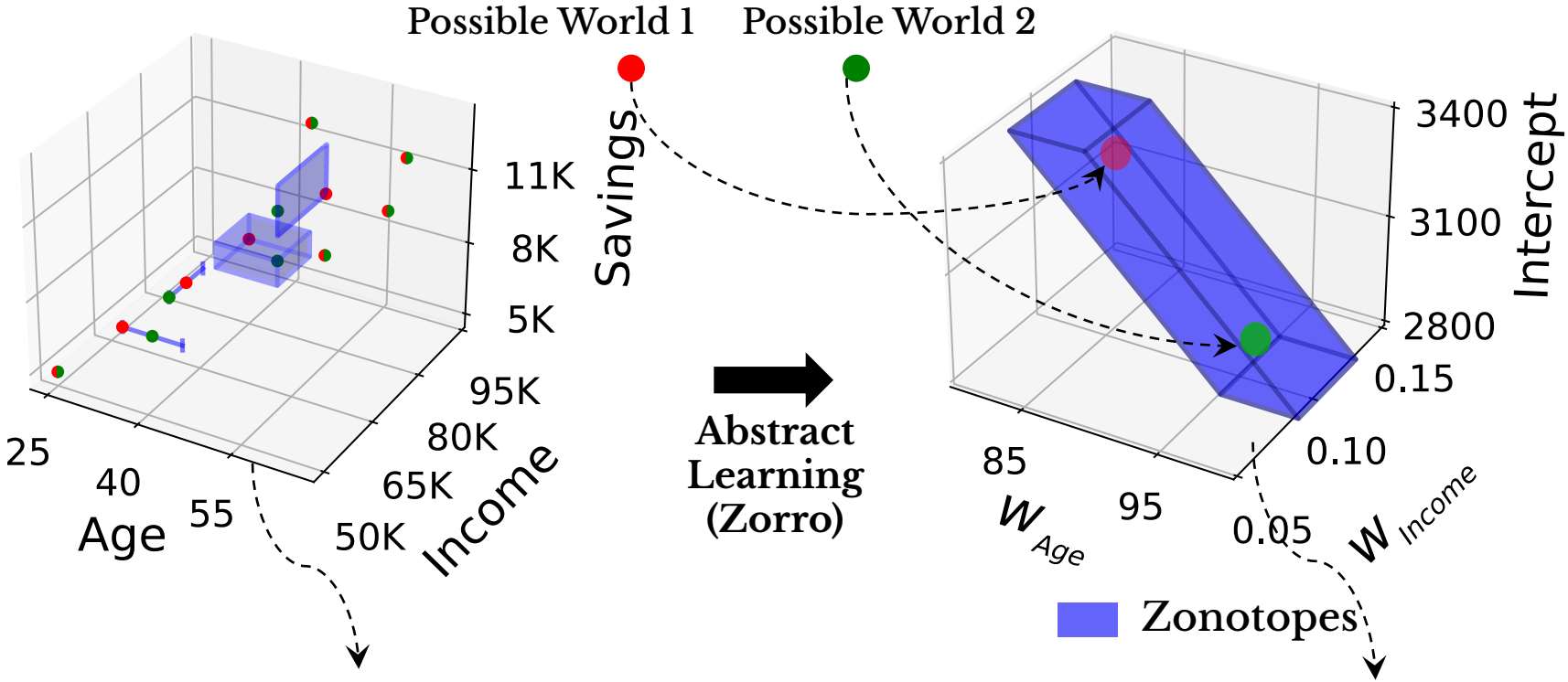
Age: 50

$Income$: $80K + 10K\ \varepsilon_1$

$Savings$: $11K + 1K\ \varepsilon_2$

$\varepsilon_1, \varepsilon_2 \in [-1, 1]$

Data uncertainty becomes model uncertainty



Age:
Income:
Savings:

$$\begin{bmatrix} 50 \\ 80K + 10K \cdot \varepsilon_1 \\ 11K + \varepsilon_2 \end{bmatrix}$$

$$, \varepsilon_1, \varepsilon_2 \in [-1, 1]$$

W_{Age}:
W_{Income}:
Intercept:

$$\begin{bmatrix} 91 + 7.7 \cdot \varepsilon_1 + 2.78 \cdot \varepsilon_3 \\ 0.1 + 0.02 \cdot \varepsilon_2 \\ 3100 - 278 \cdot \varepsilon_1 + 0.08 \cdot \varepsilon_3 \end{bmatrix}$$

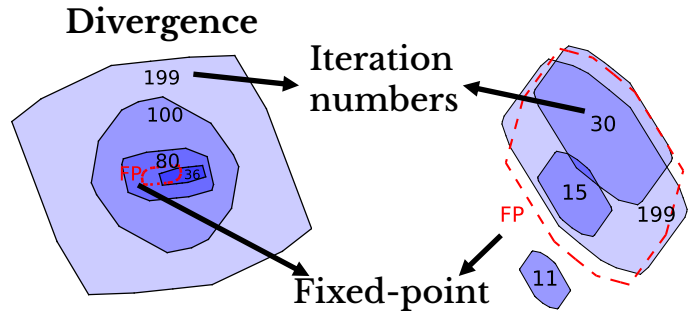
ZORRO: challenges and contributions

The challenges

- Training with uncertain values makes **symbolic expressions grow**.
- Accumulating error makes **enclosures too loose**.

Our contributions

- Control approximation and representation size to keep **updates compact**.
- For the analyzed ridge-regression setting, compute a **stable enclosure** of models across represented repairs.

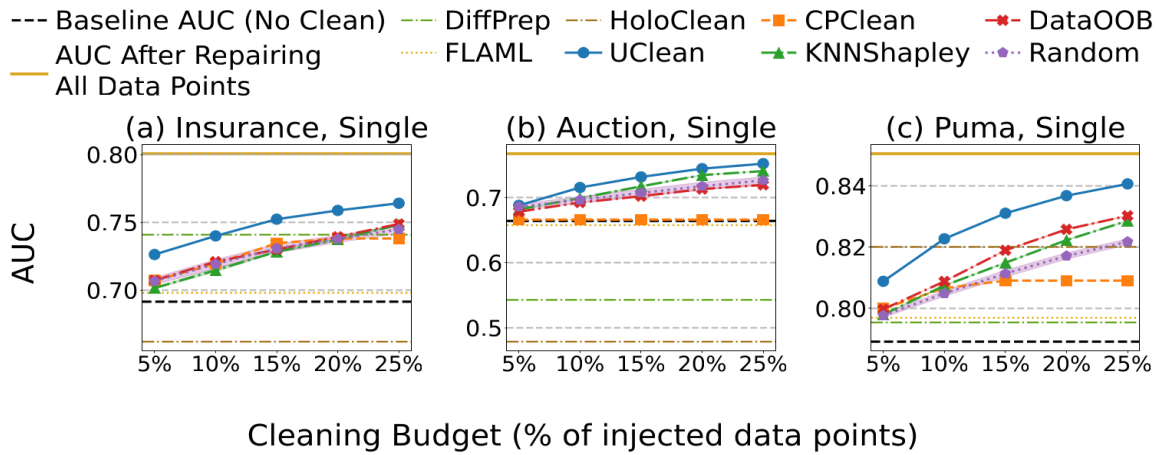


WHAT THE ENCLOSURE ENABLES

1. Establish a range of possible predictions.
2. Check whether a decision survives every represented repair.
3. Follow-up: prioritize cells for cleaning using loss sensitivity.

Targeted cleaning under a limited budget

UClean uses ZORRO loss sensitivity to prioritize uncertain cells for repair.



x: cleaning budget (% of injected data points). y: test ROC-AUC (0–1); higher means better class discrimination.

At 25%, UClean leads the plotted budgeted methods. Full oracle repair and non-oracle full-dataset cleaners are separate references.

DiffPrep, HoloClean and FLAML clean the full dataset without the ground-truth oracle.

For LLMs, first identify the training stage

Question	Input and stage	Output
Attribution	Fixed model + training candidates + behavior query	Sensitivity ranking
Selection	Candidate instructions + target development data	Subset for future training
RAG evidence	Current query + retrieved context at inference	Answer grounded in that context

- Pretraining learns from broad token sequences.
- Instruction tuning adapts weights using instruction–response examples.
- Retrieved context is an inference input; it need not update weights.

Attribution: which training data affected this behavior?

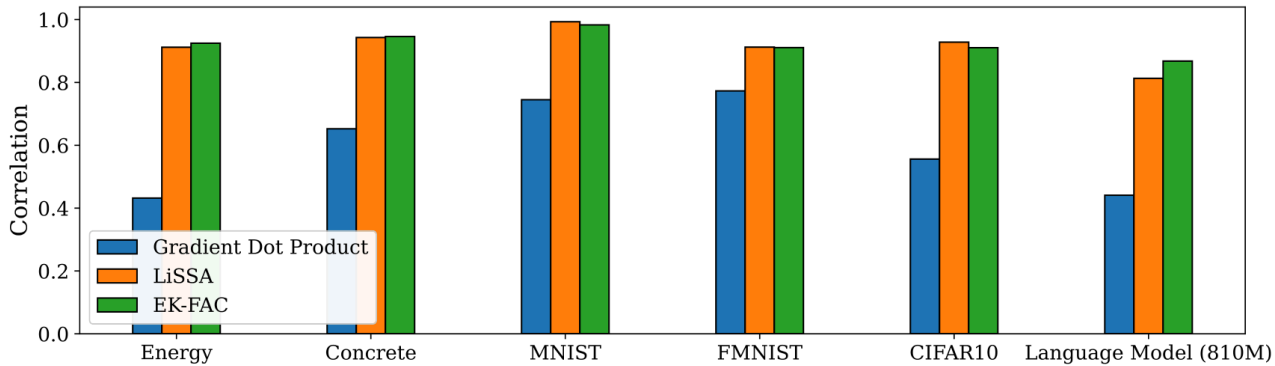
Grosse et al. · technical report · 2023

Component	Specified object
Behavior query	Prefix and completion loss at a fixed checkpoint
Candidate	A pretraining sequence or sequence segment
Computation	Query/candidate gradients and approximate inverse curvature
Result	Ranked local completion-loss sensitivity

- Grosse and colleagues study training-example influence on language-model behavior.
- Query prefix and completion + candidate training sequences → gradients and approximate curvature.
- The output ranks local sensitivity of the specified completion loss.

Read the validation target carefully

Grosse et al. · 2023 technical report



- The plotted target is proximal Bregman response, not exact leave-one-out retraining.
- Bars show Pearson correlation with PBRF; the language-model panel uses 810M parameters.
- A proximity-penalized response is useful when ordinary refitting is unstable.

A plausible document is not yet an intervention test

Test component	What must be observed
Prediction	A defined change in behavior or loss
Intervention	Remove or reweight the declared data unit
Comparison	Suitable random or matched controls and repeated seeds
Evaluation	Reserved queries and the stated outcome

- Inspect what the score predicts: a loss change, a response change or a similarity.
- Check whether the proposed data intervention realizes that prediction.
- Keep attribution separate from memorization and entitlement.

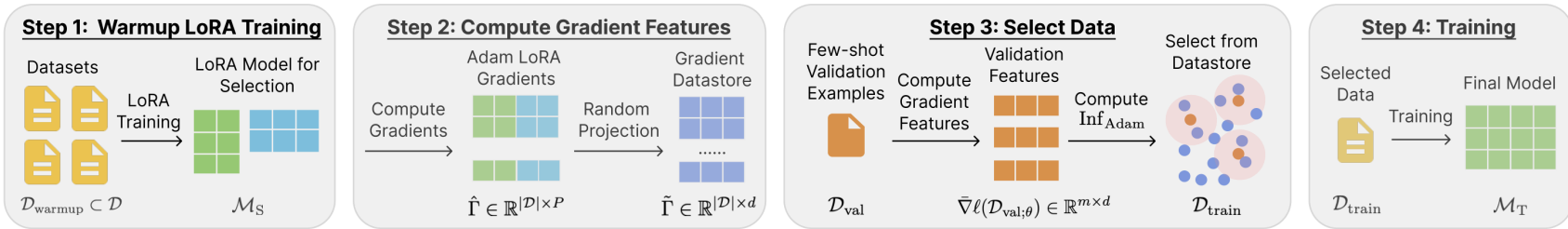
Selection: which instructions should we train on next?

Xia, Malladi et al. · ICML 2024

- A broad instruction pool may contain many examples irrelevant to a target capability.
- LESS selects examples using target-validation gradient similarity.
- The selected subset is then used in a separate fine-tuning run.

LESS builds reusable gradient features

Xia, Malladi et al. · LESS · ICML 2024 · Original four-step pipeline



- Warm up a LoRA selection model on sampled candidate instructions.
- Build projected, Adam-aware candidate update features.
- Compare target-validation gradients with candidate features to select a budgeted subset.
- Train the target model on that subset; keep its test evaluation separate.

Illustration: compare two update directions

Candidate gradient	Dot product with target (1, 0)	Cosine similarity
(1, 0)	1	1
(0, 1)	0	0

$$\Delta \ell_{\text{target}} \approx -\eta \, g_{\text{target}}^{\top} g_{\text{candidate}}$$

η > 0: small learning rate · g: gradient · illustrative plain-gradient step, not the complete LESS estimator

- Let the target-loss gradient be (1, 0).
- Candidate gradients (1, 0) and (0, 1) have cosine similarities 1 and 0.
- A small negative-gradient step from the aligned candidate reduces target loss to first order.

Selection data are not the final test

Phase	Examples allowed	What they decide
Warmup / feature construction	Candidate-pool sample	Selection representations
Subset choice	Target-development examples	Which candidates to keep
Final evaluation	Untouched target test	Whether adaptation generalizes

- Target-validation examples guide which training examples are selected.
- The final test evaluates the model after selected-subset training.
- Compare equal budgets and count warmup, scoring and training costs.

Five percent can help—but not on every task

LESS · ICML 2024

	MMLU				TyDiQA				BBH			
Data percentage	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)
LLAMA-2-7B	*51.6*	46.5 (0.5)	-	**50.2** (0.5)	*54.0*	52.7 (0.4)	-	**56.2** (0.7)	*43.2*	38.9 (0.5)	-	**41.5** (0.6)
LLAMA-2-13B	*54.5*	53.4 (0.1)	**54.6** (0.3)	54.0 (0.7)	*54.3*	53.0 (1.3)	**57.5** (0.8)	54.6 (0.3)	*50.8*	47.0 (1.6)	49.9 (0.5)	**50.6** (0.6)
MISTRAL-7B	*60.4*	60.0 (0.1)	60.6 (0.3)	**61.8** (0.4)	*57.7*	56.9 (0.2)	**61.7** (1.7)	60.3 (2.4)	*53.0*	54.5 (0.1)	**56.0** (0.8)	**56.0** (1.0)

- For Llama-2-7B, LESS at 5% gets 50.2% MMLU accuracy versus 51.6% for full data.
- TyDiQA gold-passage macro F1 is 56.2 versus 54.0 on a 0–100 scale.
- Parentheses report standard deviation across 3 training seeds.

Disentangle representation from selection

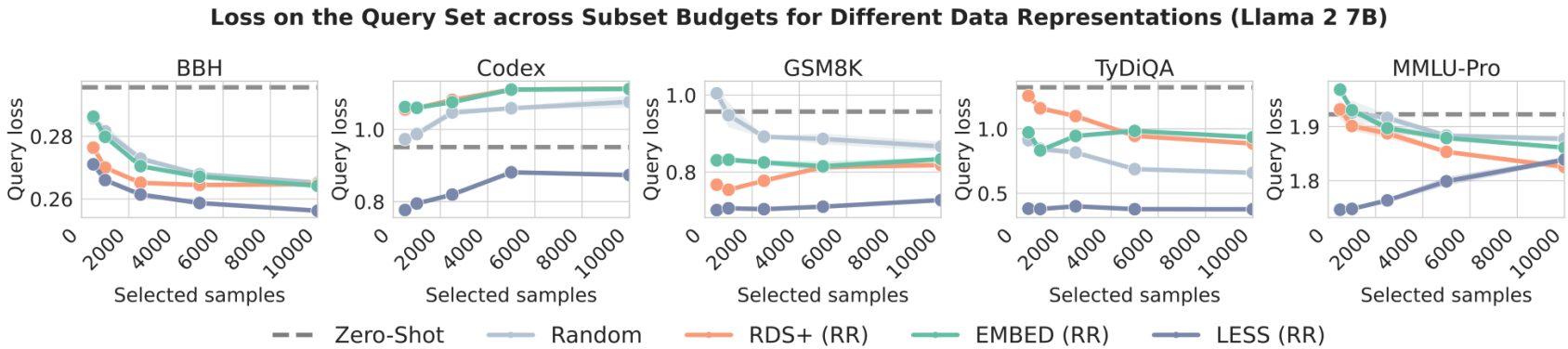
Nayak et al. · ICML 2026

Component	Controlled comparison
Representation	Gradient features versus embedding alternatives
Selection rule	Hold greedy round-robin fixed in the next plots
Budget	Number of candidate examples selected
Evaluation	Query loss versus held-out task performance

- A scorer and a subset-selection algorithm are different components.
- Nayak and colleagues compare representations under controlled selection rules.
- They vary target tasks, candidate pools and budgets.

A proxy result: loss on the query set

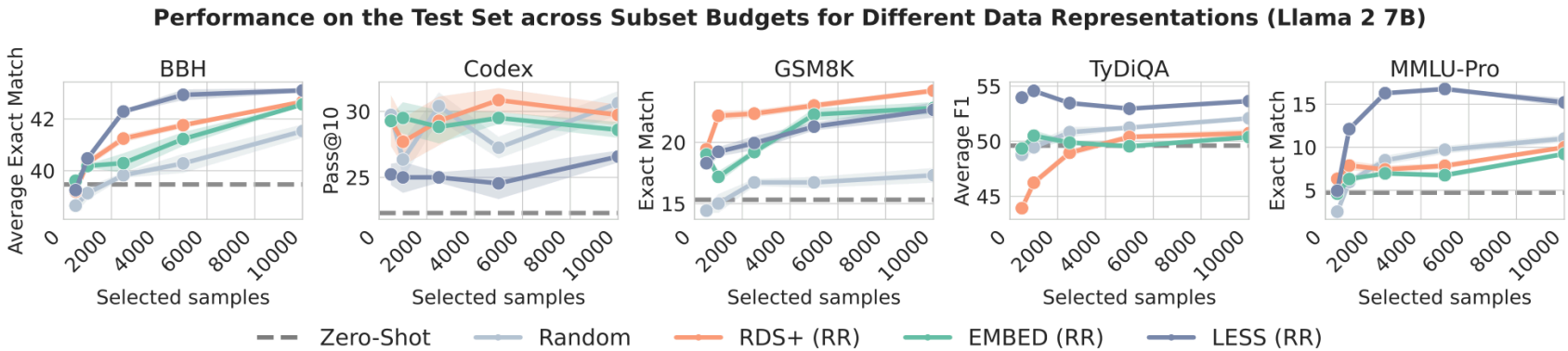
A Critical Look at Targeted Instruction Selection · ICML 2026



- Compare query loss across budgets while holding the selection rule fixed.
- Gradient representations provide a useful query-loss signal in these settings.
- Query loss is not the same outcome as held-out task performance.

The final task result depends on the budget

Nayak et al. · ICML 2026



- Compare held-out performance; shaded bands are standard errors across 3 seeds.
- Gradient-based choices often help at low budgets; gains shrink at larger budgets.
- No single method dominates all tested settings.

Choose the question before choosing the score

Decision	Relevant question	Validation target
Removal	What changes without this player?	Actual refit outcome
Coalition credit	What does the player add across contexts?	The specified cooperative game
Future selection	Which subset should train the next model?	Held-out performance after training

- Remove a player: use the full-data intervention or a validated approximation.
- Allocate coalition credit: specify the game and estimate Shapley carefully.
- Select future training data: test the chosen subset after training.

Transfer exercise: adapt an assistant responsibly

- You have a pretrained model, 100,000 candidate instructions and a 5,000-example training budget.
- The target population includes common requests and a rare safety-critical group.
- Design the selection, comparison and evaluation before choosing a method.

Feedback: make the intervention testable

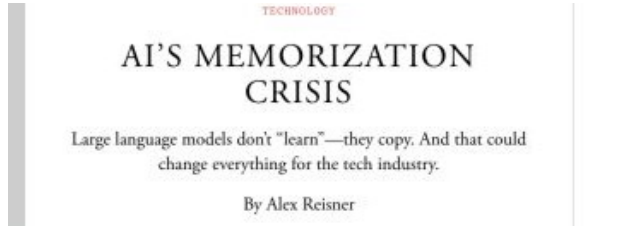
Design choice	Defensible baseline or check
Data budget	Same 5,000-example budget for random selection
Adaptation benefit	Compare with the unchanged base model
Generalization	Untouched final test, not the selection set
Responsibility	Rare-group outcomes, cost and uncertainty

- Use target-development data for selection and keep a separate final test.
- Compare selected data with random selection, the base model and feasible full-data training.
- Report overall and group outcomes, cost and uncertainty.

Reliability when language models use data

Training data, adaptation data and agent feedback each create opportunities to intervene.

- So far: how data selection and quality affect learning. Now: how language models use data, and where we can intervene.
- I will focus on three concrete failure modes and the interventions they motivate.
- Memorization: a model can reproduce sensitive or copyrighted training text.
- Forgetting: fine-tuning can weaken existing knowledge and safety behavior.
- Agent reliability: repeated runs can share the same blind spots.



TECHNOLOGY

AI'S MEMORIZATION CRISIS

Large language models don't "learn"—they copy. And that could change everything for the tech industry.

By Alex Reisner

BUSINESS

The New York Times sues OpenAI and Microsoft for using its stories to train chatbots

FINE-TUNING ALIGNED LANGUAGE MODELS COMPROMISES SAFETY, EVEN WHEN USERS DO NOT INTEND TO!

Lawyers fined for filing bogus case law created by ChatGPT

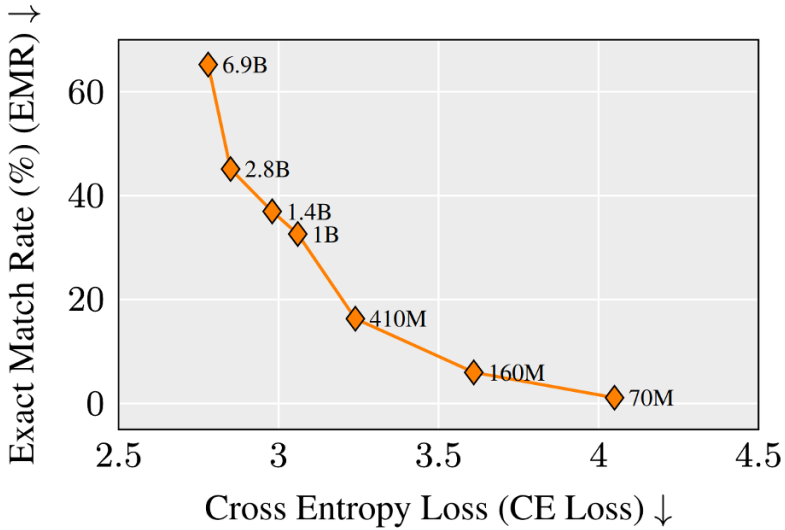
June 23, 2023 / 11:44 AM EDT / AP

Next: reducing verbatim recall, preserving knowledge, and building complementary agents.

Clippings: The Atlantic, Jan. 2026 · AP, Dec. 2023 · Qi et al., ICLR 2024 (arXiv:2310.03693) · CBS News / AP, June 2023.

Verbatim memorization in language models

Language models can reproduce training passages word for word.



Larger Pythia models predict better, yet copy more training continuations exactly.

- **Next-token training** rewards the original next token for each prefix. Fitting passage-specific patterns can lower loss. Repeated passages get more weight; rare passages can also be retained.
- **Memorization takes different forms:** reproducing a training passage word for word, or recalling its specific content in different words. Different settings call for different definitions and measures.
- Here we focus on verbatim and near-verbatim reproduction of training passages, motivated by **privacy and copyright risks**.

Figure: Prashant, Ponkshe & Salimi, TokenSwap, NeurIPS 2025. Background: Carlini et al., ICLR 2023; Feldman, STOC 2020.

Existing defenses, and the access they need

The required access determines who can use a defense.

- Pre-training defenses — deduplication, differential privacy, Goldfish — require control over the training data or the training process.
  [Kandpal, Wallace & Raffel, ICML 2022] [Abadi et al., CCS 2016] [Hans et al., Goldfish, NeurIPS 2024]
- Unlearning and fine-tuning edit the weights, stay vulnerable to extraction, and can erase capabilities and safety guardrails.
  [Sakarvadia et al., Balanced Subnet, ICLR 2025] [Russinovich & Salem, Obliviate, 2025] [Shumailov et al., 2024]
- Inference-time methods such as MemFree and CP-Fuse avoid retraining but still need the training corpus or a second model trained on disjoint data.
  [Ippolito et al., MemFree, INLG 2023] [Abad et al., CP-Fuse, ICLR 2025]
- Our setting: next-token probabilities and control over which token comes next. No weight updates, retraining or knowledge of the memorized text.

The team running the model can change token selection without changing its weights.

Access comparison: Prashant, Ponkshe & Salimi, NeurIPS 2025, Table 1.

RESEARCH PAPER

TokenSwap

TOKENSWAP: A Lightweight Method to Disrupt Memorized Sequences in LLMs

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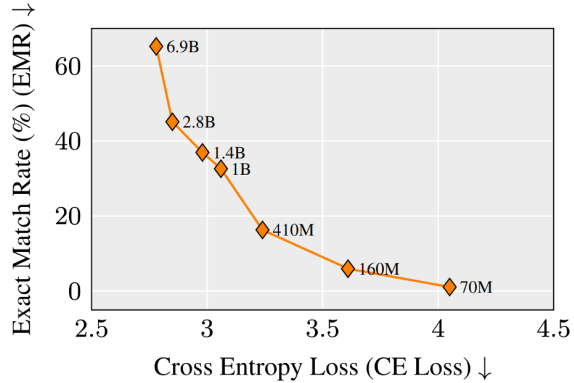
NeurIPS 2025 · Spotlight

Can small-model guidance reduce verbatim recall during generation?

TokenSwap: swap probabilities for function words

Can we reduce exact copying while retaining the large model's prediction quality?

- Smaller models **copy less training text**, but larger models predict better.
- **Small models still handle common grammar.** We use the small model to guide **function words** such as "the," "of," and "to," while leaving other tokens to the large model.

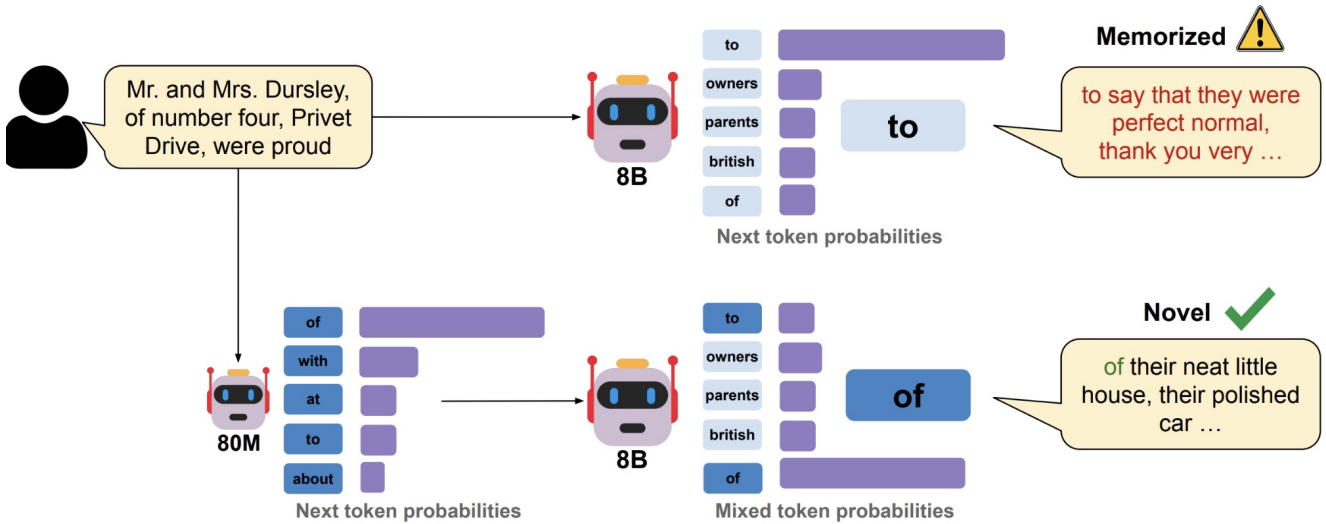


- Both models read the same text so far. We keep the large model's total probability assigned to function words, but use the **small model's preferences** to divide it among them. **All other token probabilities stay unchanged.**
- We generate the next token and repeat. A different function word **changes the context** for later predictions, which can steer generation away from a memorized sequence.

Prashant, Ponkshe & Salimi. TokenSwap, NeurIPS 2025. Section 3, Algorithm 1 and Figure 3 (standard decoding).

TokenSwap at a glance

Standard decoding on top; TokenSwap below.



Illustrative decoding step. Top: the main model follows the memorized continuation. Bottom: rescaled helper probabilities change “to” to “of”, altering the subsequent context.

A different grammar token can divert generation from an exact continuation.

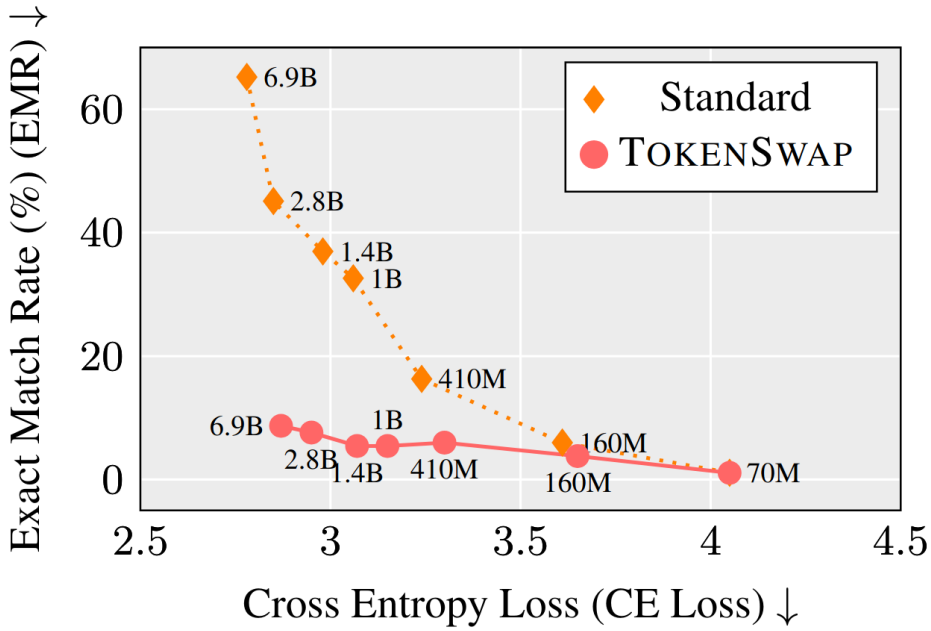
Prashant, Ponkshe, Salimi. TokenSwap. NeurIPS 2025. Figure 2.

Less copying, little change in measured quality

TokenSwap cuts exact copying by 87% (65.2% to 8.7%) for Pythia-6.9B, with only a small increase in prediction loss (2.77 to 2.87).

Separate Llama-3-8B tests retain 71.87% average accuracy on commonsense reasoning and question answering (e.g., BoolQ, PIQA and ARC).

Multi-turn conversation and instruction following also remain comparable: 7.75 vs. 7.78 on MT-Bench.



Prashant, Ponkshe, Salimi. TokenSwap. NeurIPS 2025. §4.2 and performance–memorization appendix.

Fine-tuning without forgetting



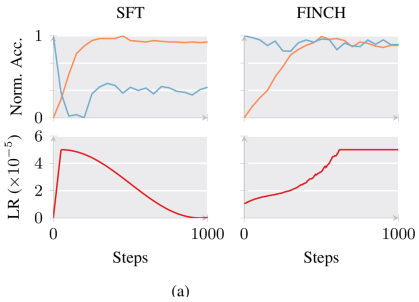


Parjanya Prajakta Prashant
Jiongli Zhu

An update can improve the new task while making the model less reliable elsewhere.

Fine-Tuning Without Forgetting via Loss-Adaptive Learning Rates

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University of California San Diego
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Qwen3-4B knowledge acquisition. Orange: new task. Blue: old tasks. Top: normalized accuracy. Bottom: learning rate.

- **Forgetting**: fine-tuning improves a new task but weakens abilities the model already had.
- Example: Qwen3-4B learns fictional authors' biographies. Accuracy rises from 53.5% to 83.5%, but general knowledge (MMLU) **falls by 19.6 percentage points**.
- Existing fixes suppress high-loss tokens or examples (FLOW, DFT, TALR, STM), but these can carry **new knowledge**.
- FINCH keeps difficult examples but takes **smaller steps on high-loss batches**, guided by a loss-based forgetting bound.

Keep the examples. Adapt the learning rate to the batch loss.

Prashant, Zhu, Creo & Salimi. FINCH, arXiv:2605.20005, 2026 (Figure 1, Table 1). Prior work: Sanyal, Wu, Lin et al., 2025.

RESEARCH PAPER

DivSkill-SQL

Residual Skill Optimization for Text-to-SQL Ensembles

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Seyedeh Baharan Khatami[1†] Canwen Xu[2] Xiaodong Yu[2] Yingyu Lin[1]
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Preprint · May 2026 · with Snowflake AI Research


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Baharan Khatami

Leaderboard

Spider 2.0-Snow | Spider 2.0-DBT | **Spider 2.0-lite**

Spider 2.0-Lite is a self-contained text-to-SQL task that includes well-prepared database metadata and documentation. This setup enables a text-in, text-out approach, facilitating faster development and evaluation. Spider 2.0-lite, which has **547** examples, is designed to handle queries for *BigQuery*, *Snowflake*, and *SQLite* databases.

Rank	Method	Score
1 (May 22, 2026)	DivSkill-SQL *Snowflake AI Research x UCSD*	73.13
2 (Apr 11, 2026)	SOMA-SQL *Oracle Cloud Infrastructure, AI Science team*	72.02
3 (Apr 29, 2026)	DecisionX Agent *DecisionX AI*	71.84

Rank 1 on Spider 2.0-Lite

How do we build agents with complementary SQL capabilities?

Text-to-SQL and the Pass@K ceiling

- Large schemas, different SQL dialects, and multi-step reasoning make one-shot generation unreliable. **Agents** inspect the schema, write and execute SQL, check results, and repair errors.
  [CHESS: Talaei et al., 2024] [ReFoRCE: Deng et al., 2025]
- CHASE-SQL uses **handcrafted strategies**, such as decomposition and query-plan reasoning, then selects among their candidates. Different strategies can recover different failures.
  [Pourreza et al., 2024] [Liu et al., TKDE 2026] [Yang et al., 2025] [Lee et al., MCS-SQL, COLING 2025]
- **Pass@K** is the chance that at least one of K candidates is correct. You cannot select a correct answer you never generated.
- Different prompts or temperatures can still repeat **the same mistakes**. More candidates help when they cover different failures.

How do we make the next attempt solve questions the others miss?

Zhu, Guan, Prashant et al. Residual Skill Optimization for Text-to-SQL Ensembles. arXiv:2605.21792, May 2026. Pass@K: Chen et al., 2021.

Skills are strategy prompts

- Generating more candidates helps only if they cover **different failures**.
- Our idea is to optimize the instructions so that each skill learns to solve **questions the other skills miss**.
- A skill is simply a **strategy prompt**. The model and tools stay the same. The instructions change.
- One prompt drafts SQL immediately and repairs errors. Another decomposes the question, **tests smaller subqueries**, and then combines them.

Zhu, Guan, Prashant et al. arXiv:2605.21792, 2026. §3.1, §3.3, Figure 1. Pairwise selection follows CHASE-SQL (Pourreza et al., 2024).

Optimize each skill on what the others miss

- Start with K skill prompts. Update each skill on questions that earlier skills still fail to solve. These are the **residual questions**.
- A prompt optimizer reads execution traces and **rewrites instructions**. Keep revisions that solve more residual questions.
- **Remove newly solved questions**, move to the next skill, and repeat across training batches.
- This targets **Pass@K**: a skill adds coverage when it succeeds where the others fail.
- **At test time**, run the learned skills to generate K SQL candidates and select one.

Zhu, Guan, Prashant et al. arXiv:2605.21792, 2026. §3.2, Algorithm 1, Example 3.2, Figure 1. Prompt optimization: Agrawal et al., arXiv:2507.19457, 2025.

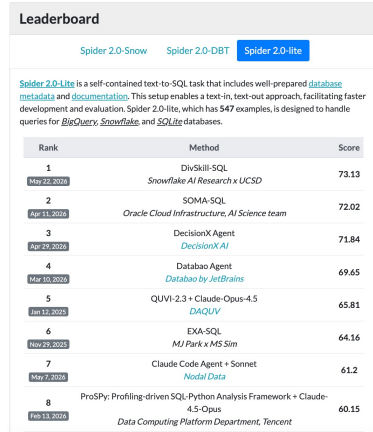
Higher selected accuracy and cross-dialect transfer

Spider2-Lite results reported in the paper, compared with CHASE-SQL.

Opus-4.6	CHASE-SQL	DivSkill-SQL	Δ
Snowflake	53.14	64.25	+11.1
BigQuery	56.59	64.88	+8.3
SQLite	63.70	64.44	+0.7

Spider2-Lite selected accuracy (%) of the returned query. Skills learned from ~200 Snowflake examples, applied to all three dialects without retraining.

- Gains appear with **both evaluated backbone models**.
- Snowflake diagnostics show **fewer invalid references** and **missing-function errors**.
- SQLite: lower Pass@8, but **higher selected accuracy**.



Leaderboard

Spider 2.0-Snow | Spider 2.0-DBT | Spider 2.0-lite

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3 (Apr 29, 2026)	DecisionX Agent *DecisionX AI*	71.84
4 (Mar 10, 2026)	Databao Agent *Databao by JetBrains*	69.65
5 (Jan 12, 2025)	QUVI-2.3 + Claude-Opus-4.5 *DAQUV*	65.81
6 (Nov 29, 2025)	EXA-SQL *MJ Park x MS Sim*	64.16
7 (May 7, 2026)	Claude Code Agent + Sonnet *Nodal Data*	61.2
8 (Feb 13, 2026)	ProSPy: Profiling-driven SQL-Python Analysis Framework + Claude-4.5-Opus *Data Computing Platform Department, Tencent*	60.15

#1 on Spider 2.0-Lite, 73.13

Leaderboard snapshot, May 22, 2026

> On Snowflake, selected accuracy improves by about **11 percentage points** over CHASE-SQL. The learned skills also transfer to the other tested dialects.

Zhu, Guan, Prashant et al. Residual Skill Optimization for Text-to-SQL Ensembles. arXiv:2605.21792, May 2026. Leaderboard: spider2-sql.github.io.

Conclusion

- Data valuation helps us choose useful examples and prioritize cleaning.
- Reasoning over possible repairs shows how data uncertainty affects models and predictions.
- Targeted interventions can reduce copying, limit forgetting, and help agents explore complementary solutions.

Trustworthy AI depends on the data models learn from and how they use it.

Valuable for the decision we can defend

Opening decision	What the lecture gives us
Inspect / repair	A ranking plus verification and a controlled audit
Retain / reweight	A defined outcome and intervention test
Reward / compensate	A specified game plus separate identity and policy rules

- Define the contribution, objective, population and baseline.
- Use the score whose question matches the intended intervention.
- Validate the action—and account for people beyond the aggregate metric.

SUPPLEMENTARY MATERIAL

Backup: evidence, ablations, and technical details

KAIROS · ZORRO · UClean · TokenSwap · FINCH · DivSkill-SQL

An RKHS is a function space built from a kernel

- A kernel represents each point by a function. For a **Gaussian kernel**, it is a bell centered at x:

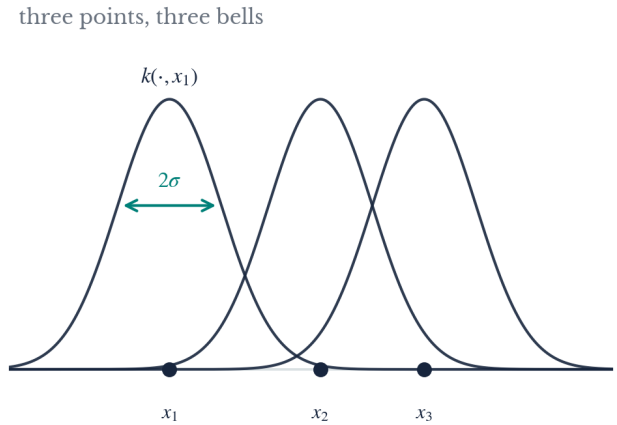
$$k(x, y) = exp(-\|x - y\|^2 / 2\sigma^2)$$

- The RKHS contains weighted sums of these functions and their **limits in the RKHS norm**. The kernel fixes their inner products:

$$f = a_1 k(\cdot, x_1) + a_2 k(\cdot, x_2) + \cdots,$$
$$\langle k(\cdot, x), k(\cdot, y) \rangle_H = k(x, y)$$

- The reproducing property expresses **function evaluation as an inner product**:

$$f(x) = \langle f, k(\cdot, x) \rangle_H$$



The kernel defines the function space, its inner product and its evaluation rule.

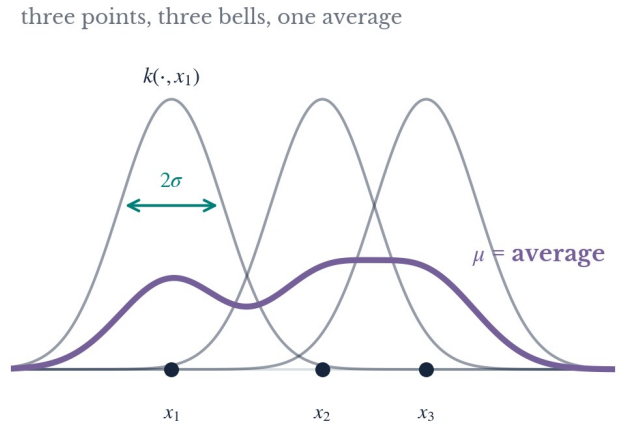
Why the RKHS choice makes MMD computable

- Averaging the reproducing property gives an inner product with the **mean embedding**:

$$E_P[f] = \langle f, \mu_P \rangle_H \quad \Rightarrow \quad E_P[f] - E_Q[f] = \langle f, \mu_P - \mu_Q \rangle_H$$

- Over functions of norm at most one, Cauchy–Schwarz gives the **distance between the mean embeddings**:

$$MMD(P, Q) = \sup_{\|f\|_H \le 1} \langle f, \mu_P - \mu_Q \rangle_H = \|\mu_P - \mu_Q\|_H$$



- Expanding the squared norm yields **three kernel averages** over pairs of samples:

$$MMD^2 = \langle \mu_P - \mu_Q, \mu_P - \mu_Q \rangle_H = E_P\, k(X, X') - 2\, E_{P,Q}\, k(X, Y) + E_Q\, k(Y, Y')$$

MMD is now a formula in kernel evaluations.

Two skills, two trajectories

Seeds from the paper, and what residual optimization added to each.

DECOMPOSE · SEED

Break complex questions into simple subqueries, build bottom-up.
1. PARSE the question into atomic requirements.
2. BUILD each piece as a standalone CTE.
3. COMPOSE the CTEs into the final query.

DECOMPOSE · WHAT OPTIMIZATION ADDED

Anchor the grain before grouping: GROUP BY exactly the output grain, no extra columns.
Ratios: totals in one CTE, subgroup counts in another, then join and divide.
Verify join keys link the intended entities — no fan-out; filter values as they appear in the data.

DIRECT CODER · SEED

You are an EFFICIENT SQL writer. Write SQL quickly, test, iterate.
1. Identify the core tables, joins, aggregations.
2. Write your best SQL IMMEDIATELY.
3. Execute; read the error, fix it. Do not over-explore.

DIRECT CODER · WHAT OPTIMIZATION ADDED

Read the schema first and find the lookup / reference tables — they almost always need a JOIN.
Map every question term to the table that owns the column; never filter on a column the table does not have.
Fix errors by type: column-not-found → join the right table; missing rows → INNER vs LEFT.

Residual feedback adds instructions that address uncovered failures.

Zhu, Guan, Prashant et al. Residual Skill Optimization for Text-to-SQL Ensembles. arXiv:2605.21792, May 2026. Examples 3.1–3.2, Appendix C (abridged).

Optimizing Pass@K, and where we differ

Related approaches optimize model parameters, prompts or agent skills.

- Some studies report that Pass@1 improvements **do not translate to higher Pass@K**, motivating attention to diversity.
- **Diversity-aware policy optimization** targets coverage through model-parameter updates.
- **Prompt optimization** proposes and evaluates instructions using task feedback. Boosted prompt ensembles also target residual errors.
- **Agent skills** encode strategies and tool-use instructions that can themselves be revised.
- DivSkill-SQL optimizes **K complementary skills** on ensemble failures without weight updates.

Move the optimization from the weights to the skills.

Zhu, Guan, Prashant et al. Residual Skill Optimization for Text-to-SQL Ensembles. arXiv:2605.21792, May 2026.

Guarantee for residual skill optimization

Proposition A.1 (Residual skill optimization approximates optimal Pass@K)**.** *Let $\mathcal{S}$ be a fixed skill family, and let $p_s(x) \in [0,1]$ denote the probability that one execution of skill s solves input x. For any skill bank $A \subseteq \mathcal{S}$, define its population Pass@K objective as*

$$F(A) = \mathbb{E}_{x\sim P}\left[1 - \prod_{s\in A}(1 - p_s(x))\right].$$

Starting from $A_0 = \emptyset$, suppose that at each round $j = 1, \ldots, K$, residual skill optimization selects

$$s_j \in \arg\max_{s\in\mathcal{S}} \mathbb{E}_{x\sim P}\left[p_s(x) \prod_{s'\in A_{j-1}} (1 - p_{s'}(x))\right],$$

where $A_{j-1} = \{s_1, \ldots, s_{j-1}\}$, and sets $A_j = A_{j-1} \cup \{s_j\}$. Let

$$A^\star \in \arg\max_{A\subseteq\mathcal{S},\ |A|\le K} F(A)$$

be the optimal size-K skill bank. Then

$$F(A_K) \ge (1 - 1/e)F(A^\star).$$

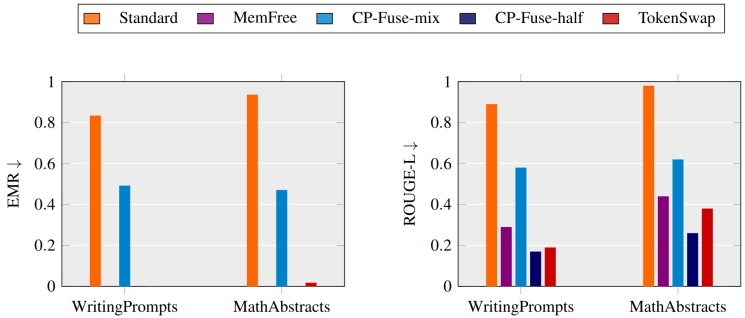
Equivalently,

$$\text{Pass@K}(\{s_1, \ldots, s_K\}) \ge (1 - 1/e) \max_{|A|\le K} \text{Pass@K}(A).$$

We show that residual skill optimization achieves at least (1 − 1/e) of the best K-skill ensemble's Pass@K.

Zhu, Guan, Prashant et al. Residual Skill Optimization for Text-to-SQL Ensembles. arXiv:2605.21792, 2026. Proposition A.1, pp. 14–15.

Memorization stress tests and separate utility checks



x: dataset. y: exact-match rate and ROUGE-L (0–1); lower is less memorization. Llama-3.2-3B, 50 epochs on 2,000 sequences.

- Exact match falls from **83.4%** to **0.1%** (WritingPrompts) and **93.6%** to **1.8%** (MathAbstracts).

Method	CE-Loss ↓	MT-Bench ↑	Commonsense (%) ↓
Standard	2.38	7.75	71.87
CP-Fuse	2.45	8.53	70.18
TokenSwap	2.43	7.78	71.87

Separate utility checks. Cross-entropy: lower is better. MT-Bench and commonsense accuracy: higher is better (the source table's downward accuracy arrow is a typo).

- Reported utility checks: cross-entropy 2.38 to 2.43, MT-Bench 7.75 to 7.78, commonsense accuracy 71.87% in both conditions.

Exact reproduction drops sharply in the stress tests. Separate utility checks show small changes and use different auxiliary configurations.

Prashant, Ponkshe, Salimi. TokenSwap: A Lightweight Method to Disrupt Memorized Sequences in LLMs. NeurIPS 2025.

Directional influence and transport objectives

The optimizer may move. Valid directional sensitivity depends on the objective and regularity.

PERTURB THE TRAINING DISTRIBUTION

$$Q_\varepsilon = (1-\varepsilon)Q + \varepsilon\delta_x$$

$$IF_d(x;P,Q) = -\lim_{\varepsilon\to 0^+} \frac{d(P,Q_\varepsilon) - d(P,Q)}{\varepsilon}$$

DIFFERENTIATE THE OPTIMIZED IPM

$$IF_{IPM} = \lim_{\varepsilon\to 0^+} \frac{E_Q[f_\varepsilon^* - f^*] - E_P[f_\varepsilon^* - f^*]}{\varepsilon} + f^*(x) - E_Q[f^*]$$

critic shift — point gap

EXACT WASSERSTEIN-1

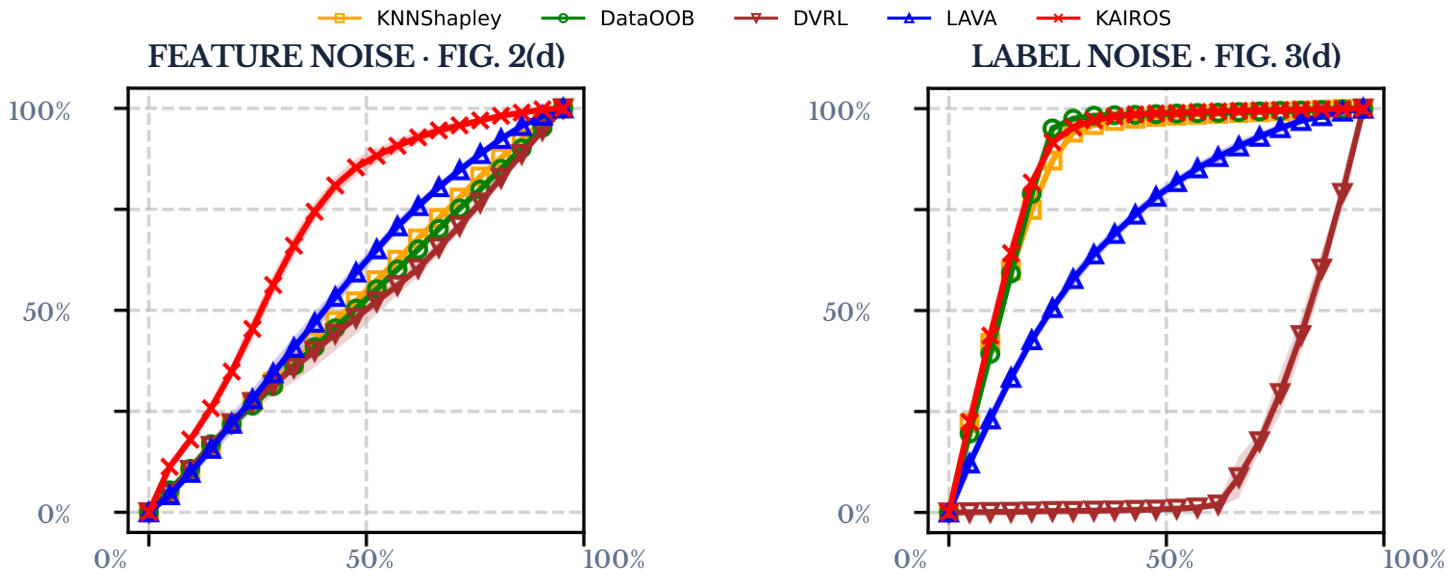
- The Kantorovich potential can be non-unique.
- Nonunique optima require care when interpreting directional sensitivities.

SINKHORN-REGULARIZED APPROXIMATION

- Regularization changes the objective and can change rankings.
- With suitable regularity, dual sensitivities are valid for that objective, but may differ from unregularized W1.

MMD yields closed-form point influence under the stated conditions.

KAIROS recovers corrupted examples in separate AG News feature- and label-noise experiments.



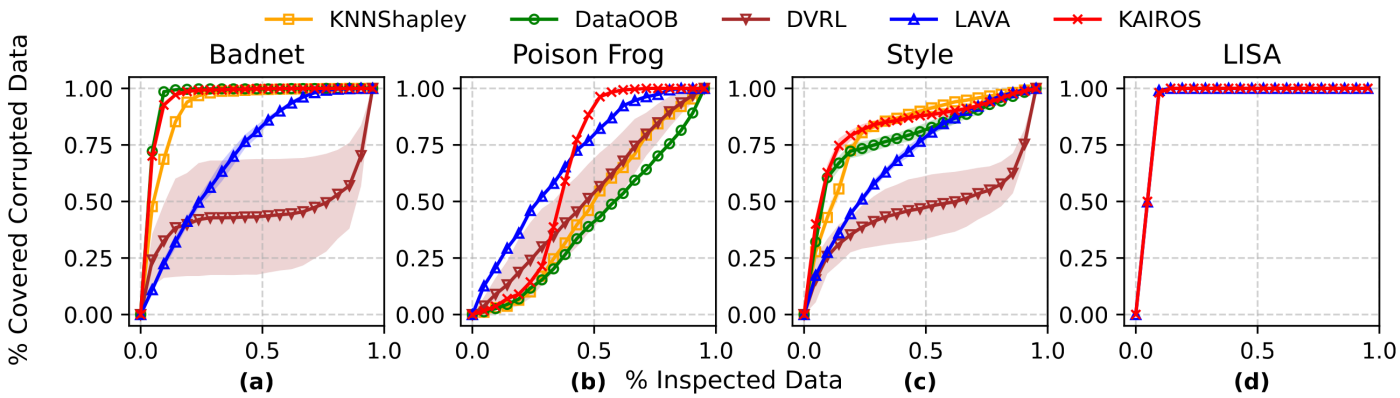
x: fraction inspected; y: fraction of corruption recovered. Inspect lowest scores first; higher curves are better.

AG News only; the two panels are separate corruption experiments, both on physical p. 7.

Across the four tested attacks, malicious points rise toward early inspection.

BadNet · Poison Frog · Style · LISA — complete tested set in Figure 4.

x: fraction of data inspected; y: fraction of malicious points recovered (0–1).
Inspect lowest scores first; higher curves recover injected attacks earlier.



In all four plotted protocols, KAIROS surfaces injected points early.
Scope: the tested Figure 4 settings—not a universal detector.

A question about learning from uncertain data.

Learning from Uncertain Data: From Possible Worlds to Possible Models

Jiongli Zhu[1] Su Feng[2] Boris Glavic[3] Babak Salimi[1]

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NeurIPS 2024



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THE QUESTION

How much can a prediction change across plausible repairs of the training data?

UNCERTAINTY-AWARE · SYMBOLIC · CERTIFIED

Possible Worlds Semantics

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Q: Total income of the three shown rows

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	30	60K	...	8K
3	35	55K	...	11K
...	...	...	...	...

→ Q(D1) = 165K

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

→ Q(D2) = 170K

...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

→ Q(D3) = 170K

...

Answer variability

- Agreement across all represented worlds gives a certain answer. Otherwise, report the range and assess whether it matters for the decision.

Shown range [165K, 170K]

Range of displayed completions

ML from Possible Repairs

Dataset with Quality Issues

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	NULL	60K	...	8K
3	35	NULL	...	[10K, 12K]
...	...	...	...	...

Replace the query with training. Each plausible repair produces a model, which predicts for the same input.

Possible Worlds

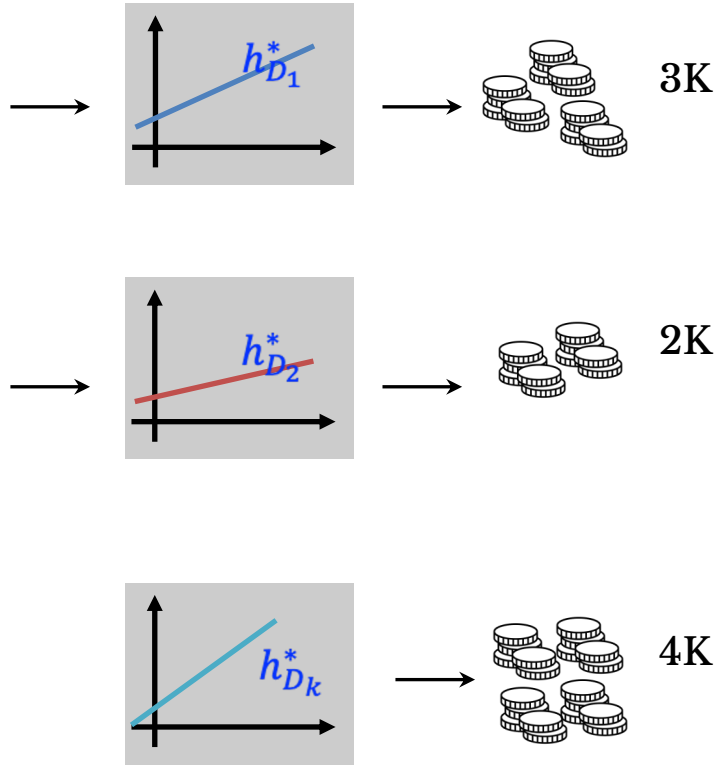
ID	Age	Income	...	Loan
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...	...	...	...	...

ID	Age	Income	...	Loan
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2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

Inference

- All represented models agree: certain prediction.
- Otherwise: a range, here 2K–4K.



Compact reasoning over possible models

- Databases represent every possible world compactly: c-tables, probabilistic databases — and relational queries run on the representation.

  [Imieliński & Lipski, JACM 1984] [Sundarmurthy et al., ICDT 2017] [Suciu, Olteanu, Ré & Koch, 2011]
- Reasoning over them is still NP-hard: certain answers are coNP-complete, prediction ranges NP-hard.

  [Abiteboul, Kanellakis & Grahe, TCS 1991] [Zhu, Feng, Glavic & Salimi, ZORRO, NeurIPS 2024]
- No support for numerical optimization: closed under relational algebra, not iterated arithmetic; symbolic gradient descent grows exponentially.
- Abstract interpretation: a representation closed under arithmetic — every training step maps the shape to a shape of fixed size, in polynomial time; sound, precise up to a little slack.

  [Cousot & Cousot, POPL 1977]

ZORRO uses zonotopes to represent uncertainty and its dependencies.

Zonotopes encode possible worlds

Dataset with Quality Issues

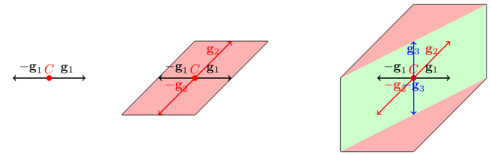
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Possible Worlds

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...	...	...	...	...

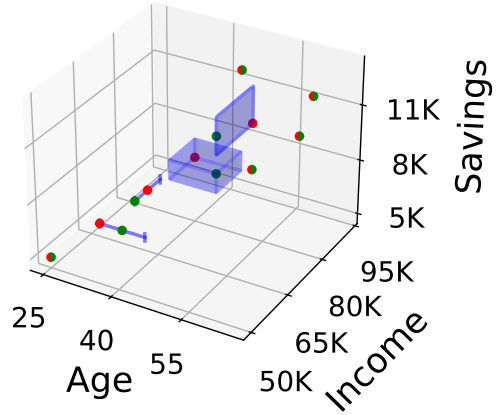
ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...

ID	Age	Income	...	Loan
1	25	50K	...	5K
2	35	60K	...	8K
3	35	60K	...	12K
...	...	...	...	...



$c + \varepsilon_1 g_1$ $\quad c + \varepsilon_1 g_1 + \varepsilon_2 g_2$ $\quad c + \varepsilon_1 g_1 + \varepsilon_2 g_2 + \varepsilon_3 g_3$

Each ε_i varies freely in $[-1, 1]$.



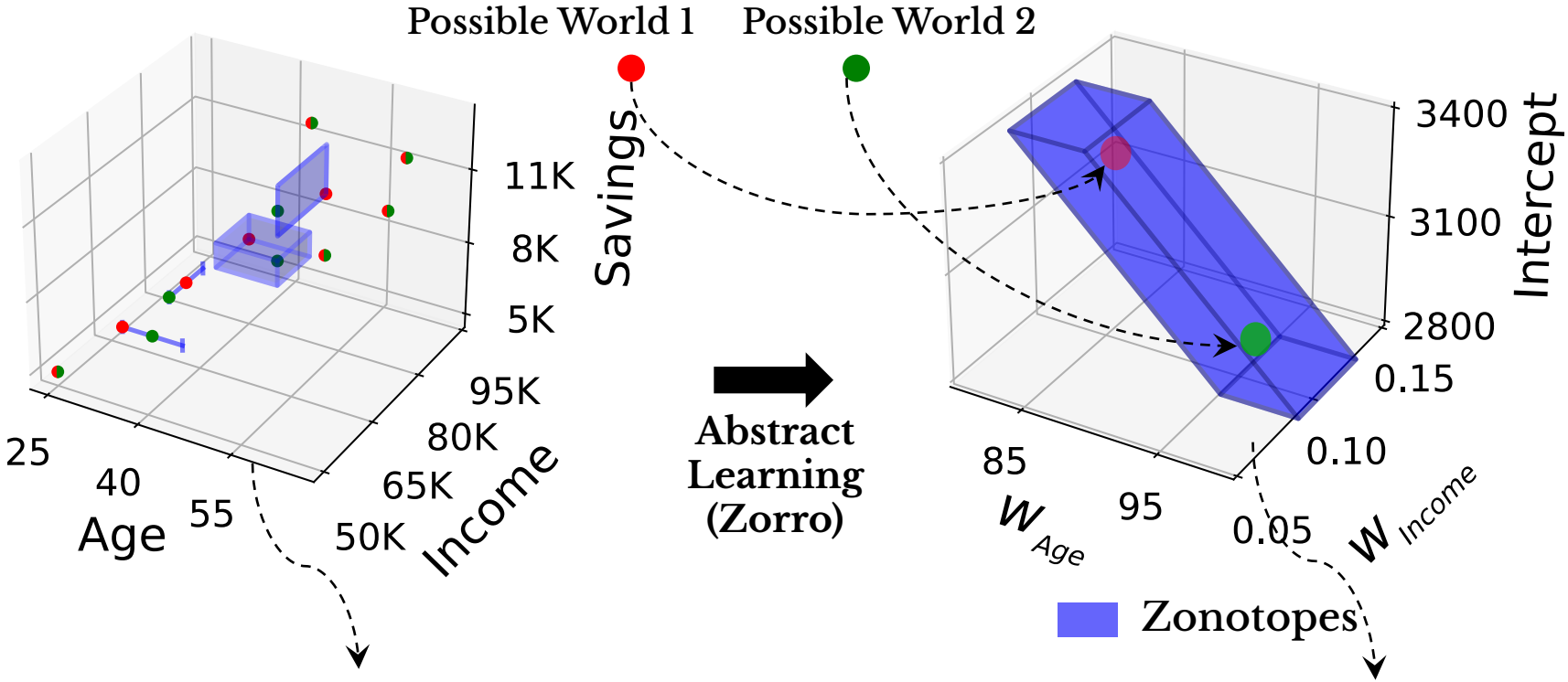
Age: 50

$Income$: $80K + 10K\ \varepsilon_1$

$Savings$: $11K + 1K\ \varepsilon_2$

$\varepsilon_1, \varepsilon_2 \in [-1, 1]$

Data uncertainty becomes model uncertainty



Age:
Income:
Savings:

$$\begin{bmatrix} 50 \\ 80K + 10K \cdot \varepsilon_1 \\ 11K + \varepsilon_2 \end{bmatrix}$$

$$, \varepsilon_1, \varepsilon_2 \in [-1, 1]$$

W_{Age}:
W_{Income}:
Intercept:

$$\begin{bmatrix} 91 + 7.7 \cdot \varepsilon_1 + 2.78 \cdot \varepsilon_3 \\ 0.1 + 0.02 \cdot \varepsilon_2 \\ 3100 - 278 \cdot \varepsilon_1 + 0.08 \cdot \varepsilon_3 \end{bmatrix}$$

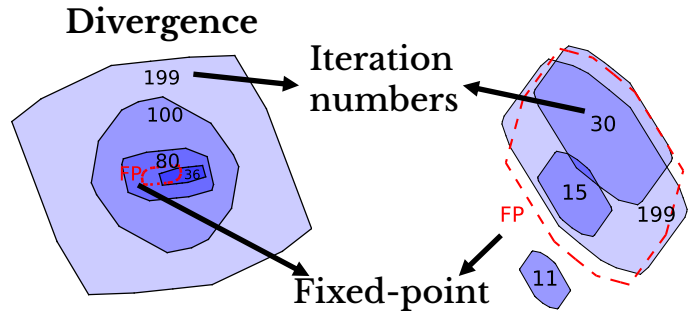
ZORRO: challenges and contributions

The challenges

- Repeated products of uncertain quantities make **symbolic expressions grow**.
- Approximation error can accumulate, making **enclosures too loose**.

Our contributions

- Bound nonlinear terms and reduce noise symbols to keep **updates compact**.
- For the analyzed ridge-regression setting, compute a **stable enclosure** of models across represented repairs.



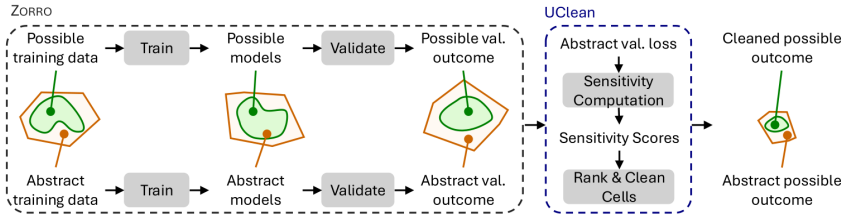
WHAT THE ENCLOSURE ENABLES

1. Establish a range of possible predictions.
2. Test whether an individual prediction or thresholded decision is stable.
3. Use downstream-loss sensitivity to prioritize uncertain cells for cleaning.

Budgeted data cleaning with ZORRO

Which uncertain cells should we clean under a limited budget?

- **Budgeted data cleaning** arises when expert review or remeasurement is costly, so we can resolve only some missing or noisy values.
- ZORRO tracks how uncertain cells affect the trained model. We extend this to express **validation loss as a function of those cells**.
- UClean ranks cells by **first-order sensitivity of this loss**, then queries an expert or trusted source to resolve the top k cells.

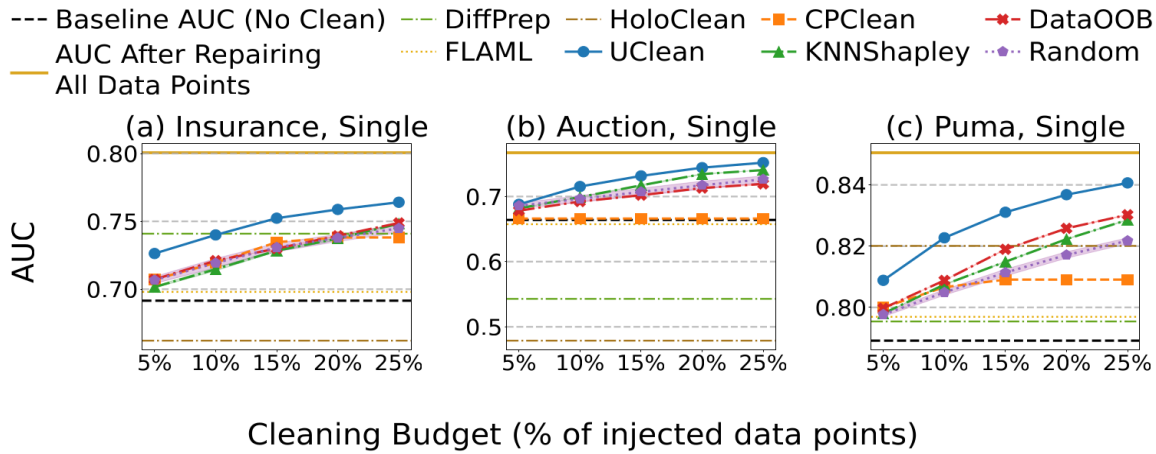


Goal: reduce uncertainty in model quality with a limited cleaning budget.

UClean: Uncertainty-Aware Data Cleaning for Machine Learning. Manuscript, 2026.

Targeted cleaning under a limited budget

Injected missingness on Insurance, Auction and Puma. Test AUC after budgeted repair.



x: cleaning budget (% of injected data points). y: test ROC-AUC (0–1); higher means better class discrimination.

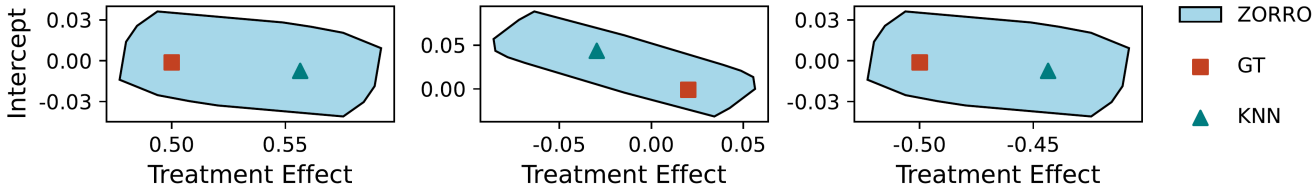
At 25%, UClean leads the plotted budgeted methods. Full oracle repair and non-oracle full-dataset cleaners are separate references.

DiffPrep, HoloClean and FLAML clean the full dataset without the ground-truth oracle.

Ground truth and KNN estimates stay inside the ZORRO causal-effect envelope.

The blue region encloses model parameters across the represented possible worlds.

LINEAR CAUSAL / RIDGE REGRESSION · FIGURE 2



x: treatment-variable coefficient. y: regression intercept. These are parameter coordinates: containment, not larger values, is the criterion.

GT squares and KNN triangles remain inside the blue envelope.

Soundness covers every represented possible world under the paper's abstraction.

Case study: linear causal model with ridge regression.

HOW TO READ THE ENVELOPE

SOUNDNESS

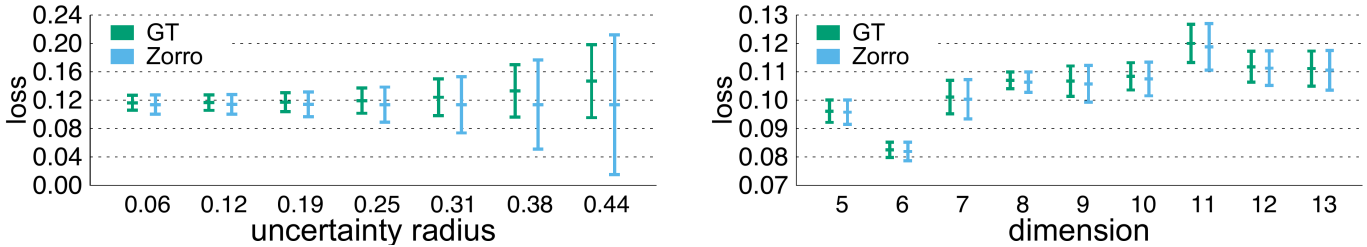
Containment is certified for the chosen abstract model.

TIGHTNESS

How close the envelope is remains empirical.

Represented worlds only—not arbitrary ambiguity.

ZORRO encloses the loss across all represented possible worlds.



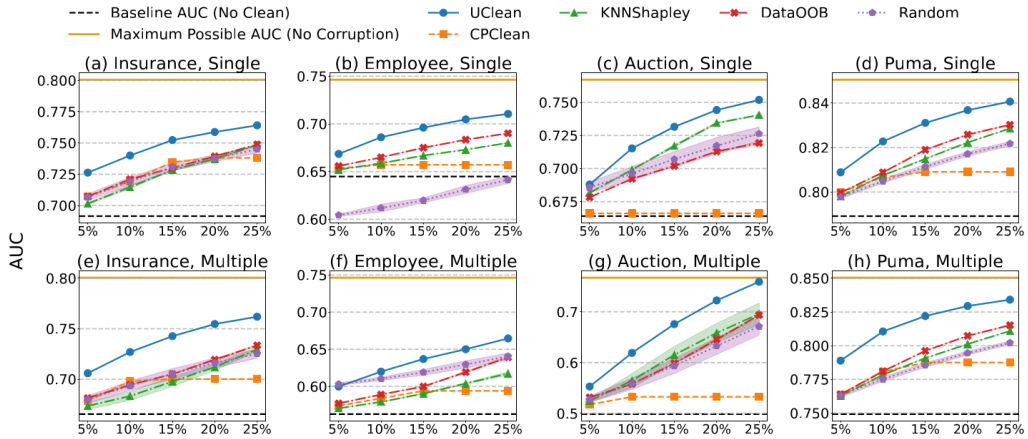
Left x: uncertainty radius. Right x: data dimension. y: loss range.
Blue: ZORRO's enclosure. Green: losses across the represented worlds.

Every enumerated represented-world loss lies inside the certified range.

Sound containment is theoretical for represented worlds; tightness is empirical and setting-dependent.

UClean recovers test AUC at smaller cleaning budgets

Four datasets × single- and multiple-column uncertainty · metric is test AUC.

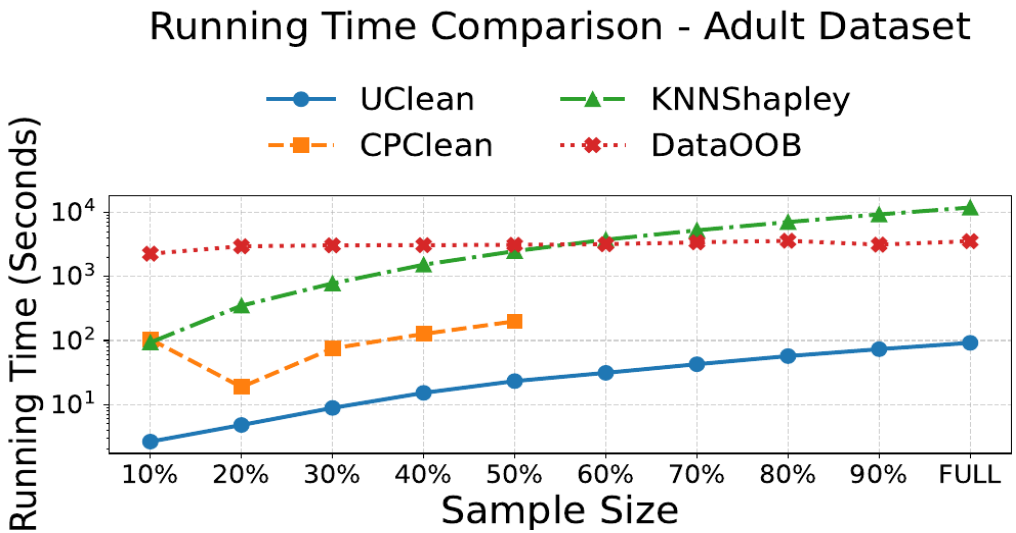


x: cleaning budget (% of injected data points). y: test AUC; higher is better at the same budget.

UClean leads the budgeted-cleaning curves in the shown panels. Gold marks the separate no-corruption reference.

Injected corruption · logistic regression · four reported datasets/settings · no global-optimality guarantee.

UClean runtime as the Adult sample grows



x: Adult dataset fraction (%; FULL = 100%).
y: wall-clock runtime in seconds, logarithmic scale; lower is faster.

2.6 s at 10% → 91.8 s at full data · reported speedups: 10×–150× versus valuation baselines.

Adult runtime study under the reported setup; runtime evidence, not cleaning-quality evidence.

TokenSwap suppresses extreme memorization across two domains.

Method	WritingPrompts				MathAbstracts			
	ML↓	EMR↓	ROUGE-L↓	Lev.↑	ML↓	EMR↓	ROUGE-L↓	Lev.↑
Standard	464.0	83.4	0.89	0.10	450.4	93.6	0.98	0.03
MemFree	17.4	0.0	0.29	0.63	6.7	0.0	0.44	0.55
CP-Fuse-mix	280.3	49.2	0.58	0.37	233.7	47.1	0.62	0.36
CP-Fuse-half	12.5	0.0	0.17	0.73	15.3	0.1	0.26	0.71
TOKENSWAP	**19.7**	**0.1**	**0.19**	**0.71**	**53.0**	**1.8**	**0.38**	**0.60**

EMR: exact-match rate (%). ROUGE-L: overlap (0–1). ML: matching length.
Lev.: normalized edit distance (0–1). Less memorization: lower EMR/ROUGE-L/ML, higher Lev.

WritingPrompts EMR 83.4 → 0.1 · MathAbstracts EMR 93.6 → 1.8

Synthetic extreme-memorization fine-tuning on Llama-3.2-3B; a stress test, not a privacy guarantee.

TokenSwap across Pythia model sizes

Table 12 Pythia sweep · a distinct protocol from the filtered 184-case Table 3 result.

SELECTED TABLE 12 ROWS

Model Size	Method	ML ↓	ROUGE-L ↑	Levenshtein ↑	EMR ↓	CE Loss↓
70M	Standard	6.57	0.180	0.709	1.09	3.95
	TOKENSWAP	**5.77**	**0.173**	**0.714**	**1.09**	**4.05**
6.9B	Standard	151.55	0.797	0.182	65.22	2.77
	TOKENSWAP	**42.65**	**0.399**	**0.536**	**8.70**	**2.87**

EMR: exact-match rate (%). ML: matching length.
ROUGE-L: overlap (0–1); lower means less memorization.
Lev.: normalized edit distance (0–1); higher means less.

70M: EMR 1.09 → 1.09 — no reduction in this row.

6.9B: stronger baseline memorization and a much larger reduction.

Table 12 · physical p. 20; exact protocol in the appendix.

6.9B · TABLE 12

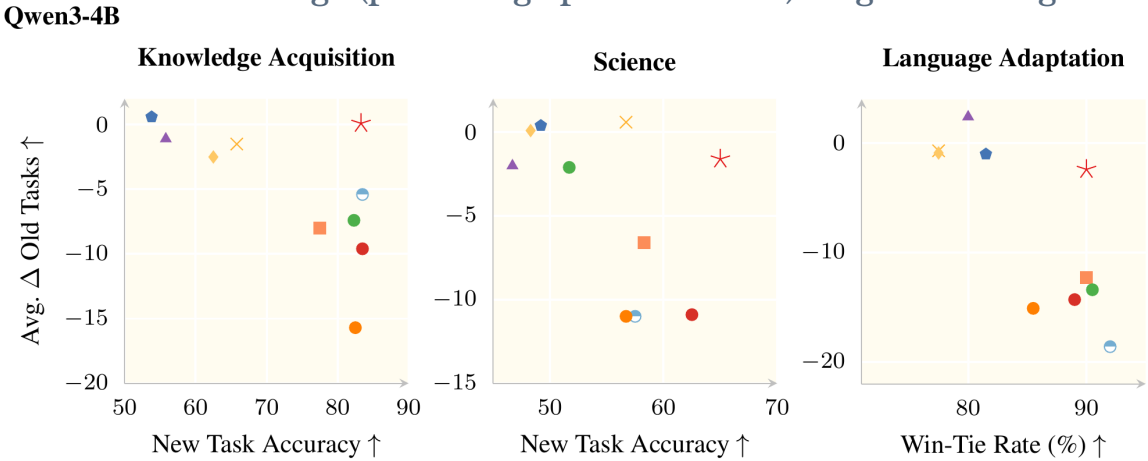
EMR 65.22 → 8.70
CE 2.77 → 2.87

SEPARATE TABLE 3 COMPARISON
Table 3's filtered 184-case result is 65.22 → 5.98.

ROUGE-L: lower means less recall. The source header's upward arrow is a typo.

FINCH balances adaptation and retention

x: new-task accuracy (%) in the first two panels; Galician win–tie rate (%) against the base in the third.
y: mean old-benchmark change (percentage points vs base). Higher and rightward are better.



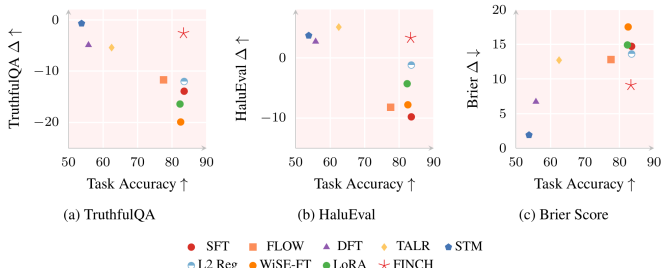
FINCH is the red ×: the reported points occupy the favorable upper-right region.

Qwen3-4B row only; the paper also reports Llama-3-8B. 'Upper-right region' is descriptive, not a formal Pareto claim.

Reliability improves, but calibration still worsens

Qwen3-4B knowledge-acquisition setting · deltas versus the pretrained base.

RELIABILITY DIAGNOSTICS · FIGURE 3



x: task accuracy (%). y: change versus the base model.
TruthfulQA/HaluEval: percentage points (higher is better).
Brier: reported score points (lower is better).

TruthfulQA: SFT −13.9 vs FINCH −2.6.

HaluEval: SFT −9.8 vs FINCH +3.3.

Brier: SFT +14.7 vs FINCH +9.1 (lower is better).

EXACT Δ VS BASE

TRUTHFULQA

−13.9 → −2.6

HALUEVAL

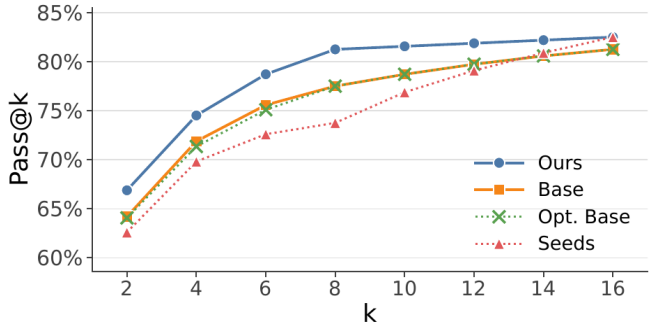
−9.8 → +3.3

CALIBRATION IS NOT FULLY PRESERVED

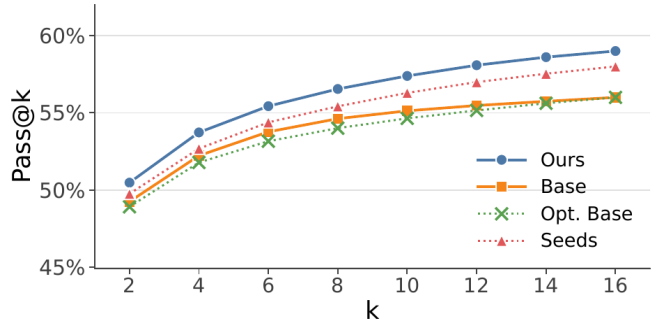
Brier still rises by +9.1.

Complementary skills improve candidate coverage

Spider 2.0-Lite



BIRD-Critic



x: generated candidates per instance (k). y: instances with at least one correct candidate (%). Higher curves mean greater candidate coverage.

On the 100-instance diagnostic subsets, baselines need 3–8 more passes to reach comparable candidate coverage.

Subset-level Pass@K diagnostic—not selected accuracy and not the full end-to-end benchmark.

Snowflake error diagnostics

Diagnostic counts from the reported Snowflake subset—not global error rates.

FOCUSED ERROR ROWS · TABLE 3

Hallucination Type	CHASE-SQL	DIVSKILL-SQL	Ratio
Pools with invalid-reference candidate	10	**7**	1.43x
Solvable pools with invalid-reference candidate	6	**2**	3.00x
Invalid-reference candidate slots	19	**13**	1.46x
Missing-function hallucination cases	6	**2**	3.00x

Solvable invalid pools decline from 6 to 2.

Missing-function errors decline from 6 to 2.

Full Table 3 · physical p. 7; counts are diagnostic, not rates.

STRONGEST REDUCTIONS

SOLVABLE INVALID POOLS

6 → 2

MISSING FUNCTIONS

6 → 2

Snowflake diagnostic counts only—not global error frequencies.

RKHS and MMD: the full derivation

How can a distance between distributions reduce to pairwise similarities?

The route through this appendix

Points become kernel features. Distributions become mean features. Their RKHS distance becomes a computable kernel average.

$$P, Q\text{: } distributions \quad X \sim P, Y \sim Q$$

k: a chosen positive-semidefinite kernel H: its function space

One construction, two views: a score gap and a feature-space distance.

Gretton et al. (2012) · Muandet et al. (2017)

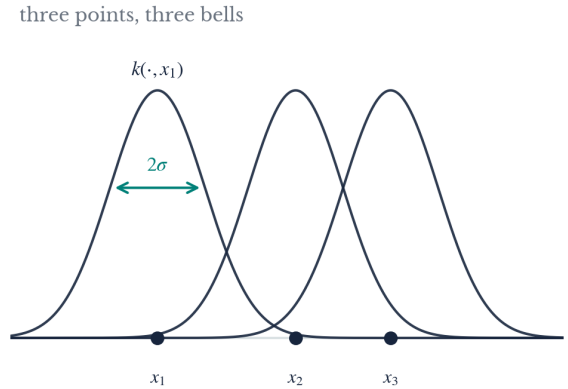
A Gaussian kernel turns a point into a bell

Fix a point x. Its feature is the whole function obtained by varying the other input.

$$k_\sigma(x, y) = exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

$$\varphi(x) = k(\cdot, x)$$

$$\langle \varphi(x), \varphi(y) \rangle_H = k(x, y)$$



σ sets the comparison scale. For this kernel, k(x,x)=1.

The kernel computes an inner product between feature functions.

Gretton et al. (2012) · Muandet et al. (2017)

The RKHS is built from kernel features

Begin with finite weighted sums of the features.

$$f = \sum_{i=1}^{n} a_i \varphi(x_i), \qquad h = \sum_{j=1}^{m} b_j \varphi(z_j)$$

$$\langle f, h \rangle_H = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j k(x_i, z_j)$$

A positive-semidefinite kernel makes squared lengths nonnegative. Completing this space adds limits in the RKHS norm.

Functions behave like vectors, with geometry determined by k.

Gretton et al. (2012) · Muandet et al. (2017)

The reproducing property evaluates a function

Use the feature centered at x as a probe for f.

$$\langle f, \varphi(x) \rangle_H = \left\langle \sum_{i=1}^{n} a_i \varphi(x_i), \varphi(x) \right\rangle_H$$

$$= \sum_{i=1}^{n} a_i \langle \varphi(x_i), \varphi(x) \rangle_H$$

$$= \sum_{i=1}^{n} a_i k(x_i, x) = f(x)$$

Evaluating f at x is an inner product with the feature of x.

Gretton et al. (2012) · Muandet et al. (2017)

The norm controls the scale of a score

$$\|f\|_H^2 = \langle f, f \rangle_H = \sum_{i,j} a_i a_j k(x_i, x_j) = a^T K a$$

Cross terms matter: the norm depends on weights and kernel similarities.

$$|f(x)| = |\langle f, \varphi(x) \rangle_H| \leq \|f\|_H \sqrt{k(x,x)}$$

For a normalized Gaussian kernel, a unit-norm function has $|f(x)| \leq 1$.

A norm cap prevents arbitrary rescaling of the average-score gap.

Gretton et al. (2012) · Muandet et al. (2017)

A distribution becomes a mean feature

Population mean and its sample version

$$\mu_P = E_{X\sim P}\varphi(X), \qquad \hat{\mu}_P = \frac{1}{n}\sum_{i=1}^{n}\varphi(x_i)$$

Average scores become inner products

$$E_P f = E_P\langle f, \varphi(X)\rangle_H = \langle f, \mu_P\rangle_H$$

Repeat the same construction for Q, using the same kernel.

The mean embedding summarizes every RKHS average score.

Gretton et al. (2012) · Muandet et al. (2017)

MMD asks for the largest average-score gap

Hold P, Q and k fixed. Choose a scoring function f from the RKHS unit ball.

$$MMD_k(P, Q) = \sup_{\|f\|_H \leq 1} \left| E_P f - E_Q f \right|$$

$$E_P f - E_Q f = \langle f, \mu_P - \mu_Q \rangle_H$$

The class is symmetric: whenever f is allowed, −f is also allowed.

The absolute and signed supremum give the same MMD.

Gretton et al. (2012) · Muandet et al. (2017)

Cauchy–Schwarz solves the supremum

$$\Delta = \mu_P - \mu_Q$$

$$|\langle f, \Delta \rangle_H| \leq \|f\|_H \|\Delta\|_H \leq \|\Delta\|_H$$

When Δ is nonzero, choose a unit vector pointing along Δ.

$$f^* = \frac{\Delta}{\|\Delta\|_H} \qquad \langle f^*, \Delta \rangle_H = \|\Delta\|_H$$

When Δ=0, every feasible score has zero gap; no normalization is needed.

Therefore MMD equals the RKHS distance between the means.

Gretton et al. (2012) · Muandet et al. (2017)

Expanding the squared distance

$$MMD_k^2(P,Q) = \left\|\mu_P - \mu_Q\right\|_H^2$$

$$= \left\langle \mu_P - \mu_Q, \mu_P - \mu_Q \right\rangle_H$$

$$= \left\langle \mu_P, \mu_P \right\rangle_H + \left\langle \mu_Q, \mu_Q \right\rangle_H$$

$$- 2\left\langle \mu_P, \mu_Q \right\rangle_H$$

Two within-distribution terms, minus twice the cross term.

Gretton et al. (2012) · Muandet et al. (2017)

The kernel trick gives three expectations

Take independent X, X′ from P and independent Y, Y′ from Q.

$$\langle \mu_P, \mu_Q \rangle_H = E_{X,Y} \langle \varphi(X), \varphi(Y) \rangle_H = E_{X,Y} k(X, Y)$$

Apply the same identity to both within-distribution terms

$$MMD_k{}^2(P, Q) = E_{X,X' \sim P} k(X, X') + E_{Y,Y' \sim Q} k(Y, Y') - 2E_{X \sim P, Y \sim Q} k(X, Y)$$

Within P + within Q − 2 × across P and Q.

Gretton et al. (2012) · Muandet et al. (2017)

The empirical distance includes self-pairs

With n points from P and m points from Q, compare the empirical mean features.

$$V_{n,m} = \left\| \hat{\mu}_P - \hat{\mu}_Q \right\|_H^2$$

$$= \frac{1}{n^2} \sum_{i,j=1}^{n} k(x_i, x_j) + \frac{1}{m^2} \sum_{i,j=1}^{m} k(y_i, y_j)$$

$$- \frac{2}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} k(x_i, y_j)$$

Nonnegative; biased for population MMD^2 because of self-pairs.

Gretton et al. (2012) · Muandet et al. (2017)

The unbiased estimator removes self-pairs

Assume independent iid samples with n,m ≥ 2. Keep all cross-sample pairs.

$$U_{n,m} = \frac{1}{n(n-1)} \sum_{i \neq j}^{n} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j}^{m} k(y_i, y_j) - \frac{2}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} k(x_i, y_j)$$

Unbiased for population MMD², but a realized value may be negative.

Gretton et al. (2012) · Muandet et al. (2017)

A complete example with two point masses

Let P put all mass at 0 and Q put all mass at a. Use a Gaussian kernel.

$$\mu_P = \varphi(0), \qquad \mu_Q = \varphi(a), \qquad k(0,0) = k(a,a) = 1$$

$$MMD_k{}^2(P,Q) = \|\varphi(0) - \varphi(a)\|_H{}^2 = 1 + 1 - 2k(0,a)$$

$$= 2\left(1 - exp\left(-\frac{a^2}{2\sigma^2}\right)\right)$$

For a=σ=1: k(0,1)=0.60653, MMD²=0.78694, MMD=0.88710.

Gretton et al. (2012) · Muandet et al. (2017)

The kernel determines what MMD detects

Linear kernel

$$k(x, y) = x^T y \quad MMD_k(P, Q) = \left\| E_P X - E_Q Y \right\|$$

Only mean differences, assuming the means exist.

Inhomogeneous quadratic kernel

$$k(x, y) = (1 + x^T y)^2$$

First and second raw moments, assuming finite second moments.

Gaussian on $\mathbb{R}^d$, $\sigma>0$: characteristic, so MMD=0 exactly when P=Q.

Sriperumbudur et al. (2010) · Muandet et al. (2017)

Bandwidth and testing answer different questions

Choose a comparison scale

For $\exp(-\text{distance}^2/(2\sigma^2))$, one median heuristic sets σ to the median nonzero pooled pairwise distance. It is a starting point.

Calibrate the observed discrepancy

For independent iid samples under P=Q, labels are exchangeable. Permute labels, recompute, and compare with the observed statistic.

Label-dependent kernel tuning must be repeated within permutations or performed on independent data.

A bandwidth chooses sensitivity; a calibrated test assesses evidence.

Gretton et al. (2012) · Muandet et al. (2017)

Computation: exact pairs or approximate features

Exact empirical computation

$$O(n^2 + m^2 + nm)\ kernel\ evaluations$$

Accumulate kernel blocks to reduce memory without changing the statistic.

Random Fourier features for shift-invariant kernels

$$k(x, y) \approx z(x)^T z(y), \qquad z(x) \in \mathbb{R}^D$$

$$V_{n,m} \approx \left\| \bar{z}_P - \bar{z}_Q \right\|^2$$

Mean-feature accumulation costs O((n+m)D), plus feature evaluation.

Gretton et al. (2012) · Rahimi & Recht (2007)

The critic gives a local data-valuation score

$$g(x) = E_P k(X, x) - E_Q k(Y, x), \qquad f^*(x) = \frac{g(x)}{MMD}$$

$$Q_\varepsilon = (1-\varepsilon)Q + \varepsilon\delta_x, \qquad \mu_{Q_\varepsilon} = \mu_Q + \varepsilon[\varphi(x) - \mu_Q]$$

$$-\frac{d}{d\varepsilon}\left\|\Delta - \varepsilon(\varphi(x) - \mu_Q)\right\|_H^2\Big|_{\varepsilon=0} = 2\langle \Delta, \varphi(x) - \mu_Q \rangle_H$$

$$= 2[g(x) - E_Q g]$$

For MMD>0: local decrease = f*(x) minus the Q-average of f*.

A point helps locally when its critic score exceeds the current Q average.

Gretton et al. (2012) · Muandet et al. (2017)

Recap: the identities and their boundaries

$$\varphi(x) = k(\cdot, x) \quad \langle f, \varphi(x) \rangle_H = f(x)$$

$$\mu_P = E_P \varphi(X) \quad E_P f = \langle f, \mu_P \rangle_H$$

$$MMD_k(P, Q) = \left\| \mu_P - \mu_Q \right\|_H$$

The plotted mean is a feature function, not generally a density.
MMD uses RKHS geometry, not the area between curves.

Population MMD, empirical squared distance, and an unbiased estimate of squared MMD are different objects.

Features → means → norm → pairwise kernels.

Gretton (2012); Muandet (2017); Sriperumbudur (2010); Rahimi & Recht (2007). Details in notes.

The reweighting path

Take n equally weighted observations, with n > 1. Keep the reference P fixed.

$$Q = \frac{1}{n}\sum_{j=1}^{n} \delta_{x_j}$$

Nudge Q toward the selected observation x_i

$$Q_\varepsilon = (1 - \varepsilon)Q + \varepsilon\delta_{x_i}$$

Selected observation x_i

$$w_i(\varepsilon) = \frac{1 - \varepsilon}{n} + \varepsilon$$

Each other observation

$$w_j(\varepsilon) = \frac{1 - \varepsilon}{n} \quad (j \neq i)$$

At ε = 0, Q is unchanged. The weights always sum to 1.
Positive ε upweights x_i and negative ε downweights it.

KAIROS, NeurIPS 2025, Definition 1. Leave-one-out algebra and Taylor expansion shown here.

Exact leave-one-out

Choose ε so the selected observation's weight becomes zero.

$$\frac{1-\varepsilon}{n} + \varepsilon = 0 \;\Rightarrow\; 1 + (n-1)\varepsilon = 0$$

$$\varepsilon_{LOO} = -\frac{1}{n-1}$$

$$Q_{\varepsilon_{LOO}} = \frac{n}{n-1}Q - \frac{1}{n-1}\delta_{x_i}$$

$$= \frac{1}{n-1}\sum_{j \neq i} \delta_{x_j} = Q_{-i}$$

The selected weight is 0. Every other observation has weight 1/(n − 1). This is exactly the leave-one-out distribution.

Example: n = 5 gives ε = −1/4, one weight of 0, and four weights of 1/4.

KAIROS, NeurIPS 2025, Definition 1. Leave-one-out algebra and Taylor expansion shown here.

Why the approximation is first order

Use the local slope to estimate the finite removal step.

$$g(\varepsilon) = d(P, Q_\varepsilon), \qquad IF_d(x_i; P, Q) = -g'(0)$$

Taylor expansion around ε = 0

$$g(\varepsilon) - g(0) = \varepsilon g'(0) + R(\varepsilon)$$

Substitute ε = −1/(n − 1), the exact removal step:

$$d(P, Q_{-i}) - d(P, Q) = \frac{IF_d(x_i; P, Q)}{n - 1} + R_i$$

First-order keeps slope × step and drops the remainder R_i.
Positive influence predicts that removal increases the discrepancy.

Assume g is differentiable across 0 and twice differentiable with $|g''| \le M$ on $[-1/(n-1), 0]$.
Then $|R_i| \le M / [2(n-1)^2]$. A smaller step improves the approximation when curvature stays controlled.
KAIROS, NeurIPS 2025, Definition 1. Leave-one-out algebra and Taylor expansion shown here.